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REGULAR CONTRIBUTION

Hybrid attention-enhanced explainable model for encrypted traffic detection and classification

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Abstract

Encrypted traffic detection and classification play a crucial role in modern network security, mainly as encryption protocols such as TLS, VPNs, and Tor become ubiquitous. This paper presents a novel approach integrating Hybrid Attention-based feature enhancement with a LightGBM classifier to improve the interpretability and performance of encrypted traffic classification. Despite advancements in machine learning (ML) and deep learning (DL) techniques, existing models face scalability, explainability, and generalization limitations across diverse network environments. To address these challenges, we develop an augmented dataset that integrates multiple publicly available encrypted traffic datasets, enhancing model robustness and diversity. Additionally, we incorporate Explainable AI (XAI) techniques, including SHAP and LIME, to analyze the importance of features and refine classification strategies. Our experimental evaluation demonstrates the proposed model's superiority in binary and multi-class classification tasks, outperforming state-of-the-art approaches. This study highlights the significance of adaptive classification models and optimized feature representations in encrypted traffic detection, setting the stage for future advancements in network security and encrypted traffic analysis.

Keywords Hybrid Attention-Based Classification · LightGBM for Encrypted Traffic · Encrypted Traffic Analysis · Feature Interpretability · Explainable AI (XAI) for Traffic Classification · Multi-Head Attention for Network Security · SHAP and LIME in Encrypted Traffic Detection · Class-Specific Threshold Optimization · Deep Learning

1 Introduction

The rapid proliferation of encrypted network traffic has significantly enhanced privacy and security while posing new challenges for network management and cybersecurity [1–4]. Encrypted traffic now constitutes the majority of internet communication, driven by the widespread adoption of encryption protocols such as Transport Layer Security (TLS), Virtual Private Networks (VPNs), and Tor [5].

While encryption safeguards user privacy, it also enables malicious actors to conceal their activities, making it crucial to develop robust methods for encrypted traffic detection and classification [2–4]. Traditional deep packet inspection (DPI) techniques are ineffective due to encryption, necessitating machine learning (ML) and deep learning (DL) approaches that leverage traffic flow characteristics and statistical features to classify encrypted traffic [5, 6].

Recent studies have explored supervised and semi-supervised learning techniques to classify encrypted traffic, focusing on statistical patterns rather than payload-based methods, which are no longer feasible with the increasing use of TLS and other encryption protocols [7]. Feature-based classification methods, including those using machine learning algorithms such as decision trees and deep neural networks, have demonstrated significant success in accurately categorizing encrypted network flows [8].

Moreover, hybrid approaches that combine multiple classification techniques have been introduced to improve detection accuracy and address dataset biases [6]. However, existing classification models still face challenges related to dataset bias, explainability, and generalization across diverse network environments [1, 9, 10]. These issues arise due to variations in network conditions, encryption protocols, and the increasing complexity of encrypted traffic patterns.

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Explainable AI (XAI) techniques are being explored to provide insights into model decisions and improve the interpretability of classification results, ensuring that ML-based encrypted traffic detection methods can be effectively deployed in real-world network environments [1].

This paper aims to address these challenges by introducing several key contributions to the field of encrypted traffic classification:

1. The **Development of an Augmented Dataset** integrating multiple publicly available datasets to improve diversity and robustness in encrypted traffic classification.
2. The **Design and Implementation of an Explainable AI-powered Classification Model**, integrating a Hybrid Attention-based feature enhancement mechanism with a LightGBM classifier for improved interpretability and performance.
3. A **Feature Importance Analysis & Model Improvement using XAI insights** using SHAP and LIME to interpret model decisions and refine classification strategies.
4. A **Comprehensive Evaluation** comparing the proposed approach against state-of-the-art classification methods to demonstrate its effectiveness in real-world network environments.

The remainder of this paper is structured as follows. Section 3 - Related Work explores various machine learning and deep learning approaches for encrypted traffic detection, including traditional ML techniques, deep neural networks, and hybrid models. Section 4 - Proposed Model details the architecture and design of the proposed system, including dataset generation, preprocessing, feature selection, and classification framework. Section 5 - Experiments & Results presents the experimental setup, dataset augmentation process, and model evaluation. Section 6 - Analysis & Discussion provides an in-depth analysis of the results, including comparisons with state-of-the-art classification models and an evaluation of optimization strategies. Finally, Section 7 - Conclusions summarizes the contributions of this work, and finally Section 8 - Challenges & Future Work outlines the challenges faced and directions for future research.

2 Motivation

The exponential growth of encrypted network traffic has significantly reshaped the digital landscape, enhancing user privacy while complicating network security and monitoring. While encryption is essential for protecting sensitive data, it also enables malicious actors to evade traditional security measures. This dual nature necessitates advanced methodologies for detecting, analyzing, and classifying encrypted traffic without compromising user privacy.

Existing encrypted traffic classification approaches face scalability, accuracy, and adaptability limitations. Traditional machine learning models struggle to keep pace with evolving encryption techniques, while deep learning models, despite their promise, demand extensive computational resources and often lack interpretability. Additionally, the scarcity of diverse and representative public datasets restricts model generalizability, limiting their effectiveness in real-world scenarios.

This work proposes a modular and scalable framework for encrypted traffic detection and classification to address these challenges. The approach aims to improve classification accuracy, scalability, and interpretability by integrating attention-based feature enhancement with hybrid machine learning models. The methodology extends beyond binary classification, facilitating fine-grained categorization of encrypted traffic into specific categories such as VPN, Tor, and others.

The proposed framework balances privacy preservation and effective security monitoring through an open-source network traffic analyzer and an advanced detection model. Designed for real-world deployment, it ensures scalability, adaptability to emerging encryption techniques, and interpretability through explainable AI methods such as SHAP and LIME. This work contributes to cybersecurity by advancing encrypted traffic analysis and setting a benchmark for future research.

3 Related work

The increasing complexity and widespread adoption of encrypted network traffic necessitate advanced classification and detection methods. This section reviews existing approaches structured according to the taxonomy in Figure 1. Research efforts are categorized into Machine Learning, Deep Learning, and Hybrid Models, with an analysis of their strengths, limitations, and contributions to network security.

Machine Learning models, such as J48, Random Forest (RF), and Instance-Based Learning (IBk), rely on handcrafted features and are computationally efficient but often struggle with complex traffic patterns. Deep Learning techniques, including CNNs with multi-head attention, Spiking Neural Networks (SNNs), Generative Adversarial Networks (GANs), and Transformers, automate feature extraction and effectively capture intricate traffic behaviors. Hybrid Models combine machine learning and deep learning to improve interpretability and performance, employing techniques such as LSTM + RF, SVM + KNN, and Multi-Layer Perceptrons (MLP) have been pivotal.

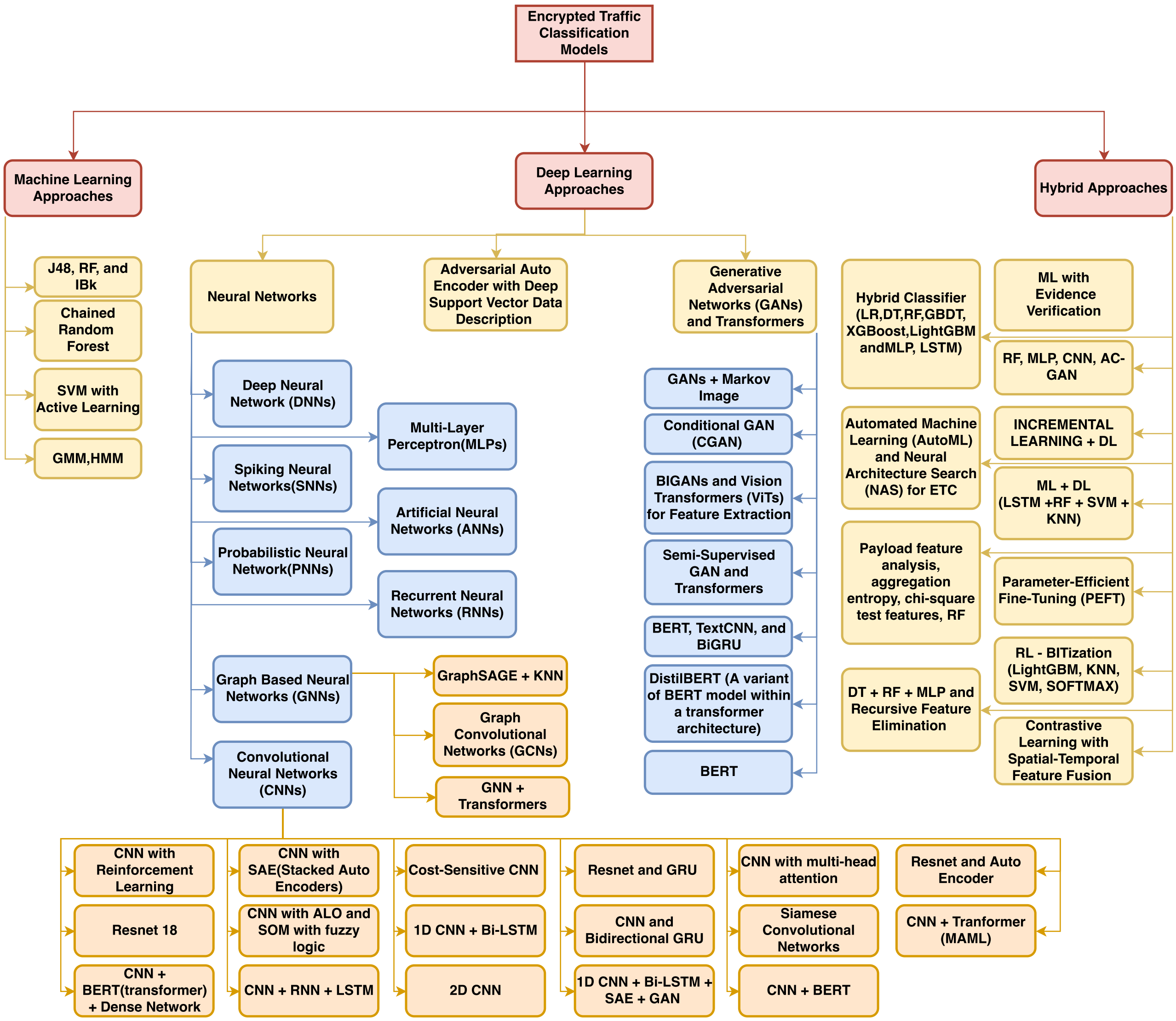

Fig. 1 Taxonomy of Encrypted Traffic Detection Models Derived from Previous Technical Literature.

This taxonomy provides a structured understanding of encrypted traffic classification methodologies, facilitating an assessment of their real-world applicability. The subsequent analysis identifies key research gaps and future directions, guiding advancements in this domain.

3.1 Machine learning models in encrypted traffic detection

Machine learning (ML) has played a pivotal role in encrypted traffic classification by leveraging statistical patterns and feature-based learning. Various classifiers, including decision trees, random forests, and ensemble methods, have been employed to enhance detection accuracy while maintaining computational efficiency.

Zaki et al. [11] propose GRAIN, a multi-label classification approach using a classifier chain with Random Forests to categorize traffic at multiple levels based on seven statistical features. The model achieves high accuracy, with F-measure scores of 99%, 93%, and 88%, demonstrating its robustness in network security applications. Shi Dong [12] addresses class imbalance in network traffic classification by introducing CMSVM, a cost-sensitive SVM with active learning, which dynamically adjusts class weights to enhance precision and recall across imbalanced datasets. Yao et al. [13] combine Gaussian Mixture Models with Hidden Markov Models (MGHMM) to improve encrypted protocol identification, outperforming existing methods in multi-dataset evaluations. Choorod et al. [14] develop a Tor traffic classification model using character statistical-based features extracted directly from encrypted payloads, achieving 95.65% accuracy without deep packet inspection.

While ML models provide scalable and interpretable solutions, their effectiveness diminishes against evolving encryption techniques and obfuscation methods. This underscores the need for hybrid models integrating ML with deep learning for improved adaptability and accuracy.

3.2 Deep learning models in encrypted traffic detection

Deep learning (DL) has significantly advanced encrypted traffic classification by enabling hierarchical feature extraction without extensive manual engineering. This section examines DL methodologies that enhance detection accuracy and adaptability, including autoencoders and multimodal architectures.

3.2.1 Autoencoders

Autoencoders (AEs) are widely used for unsupervised feature learning and anomaly detection in encrypted traffic classification. These models encode traffic patterns into lower-dimensional representations, facilitating efficient detection while minimizing reconstruction errors.

Lv et al. [15] propose AAE-DSVDD, a one-class classification model integrating Adversarial AutoEncoders (AAE) with Deep Support Vector Data Description (DSVDD) to mitigate hypersphere collapse. Evaluations on the ISCXVPN dataset demonstrate its superiority over traditional one-class models such as OCSVM, iForest, and GANomaly, achieving higher AUC scores across multiple traffic types. Aceto et al. [16] introduce DISTILLER, a multimodal multitask classifier for encrypted traffic. Unlike conventional models relying on single input modalities, DISTILLER extracts intra- and inter-modality dependencies while performing multiple classification tasks. A two-phase training approach-pre-training followed by fine-tuning-enhances feature learning and prevents modality dominance. The ISCX VPN-nonVPN dataset results show superior classification accuracy with reduced computational overhead.

Autoencoders provide a practical approach for unsupervised encrypted traffic classification, optimizing feature representation and anomaly detection. However, their reliance on reconstruction-based learning limits adaptability to dynamic traffic patterns, highlighting the need for hybrid approaches.

3.2.2 Neural networks

Neural networks have significantly advanced encrypted traffic classification by enabling sophisticated feature extraction and adaptive learning [17, 18]. Various architectures have been developed to enhance accuracy and efficiency, addressing key challenges in network security.

Jorgensen et al. [19] introduce Probabilistic Neural Networks (PNNs) with Uncertainty Quantification (UQ) for VPN application labeling. Their model employs prototypical networks with Mahalanobis distance-based out-of-distribution detection, achieving an F1-score of 0.98 by effectively distinguishing predictive and model uncertainty. Song et al. [20] propose I^2RNN, an Incremental and Interpretable Recurrent Neural Network that enables adaptive encrypted traffic classification without full retraining. It enhances interpretability by ranking time-series features and computing inter-class distances.

Zhou et al. [21] integrate entropy estimation with deep neural networks (DNNs) for application-level classification, improving precision by up to 7 percentage points over traditional ML techniques. Pathmaperuma et al. [22] develop a DNN framework for fine-grained in-app activity classification, incorporating probability-based filtering to improve robustness against unseen traffic patterns. Their model achieves over 90% accuracy for trained in-app activities and 79% for filtering unknown activity data.

Rasteh et al. [23] explore Spiking Neural Networks (SNNs) with Surrogate Gradient Learning for encrypted traffic classification, capturing packet size distributions and inter-arrival times while maintaining temporal correlations, achieving 95.9% accuracy on the ISCX VPN-nonVPN dataset. Xu et al. [24] introduce FastTraffic, a lightweight deep learning framework optimized for real-time encrypted traffic classification on resource-constrained devices. Utilizing an MLP with N-gram feature embedding, FastTraffic balances computational efficiency and accuracy, achieving superior classification with a model size of only 0.43M parameters.

Neural network-based approaches continue to improve encrypted traffic classification, offering enhanced adaptability and efficiency. However, real-time deployment and generalization to novel traffic patterns remain key challenges, necessitating further optimization.

Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs) have become central to encrypted traffic classification, leveraging hierarchical feature extraction to capture complex traffic patterns [25–27]. Their ability to model spatial dependencies in encrypted data has significantly improved classification accuracy and adaptability.

He et al. [28] propose an image-based approach that transforms network session data into grayscale images, processed by a 1D-CNN, achieving an F1-score of 98.64% for encrypted traffic and 99.55% for VPN classification. Moreira et al. [29] integrate CNNs with reinforcement learning (RL) using a Deep Q-Network (DQN) agent to dynamically adjust packet sampling rates, achieving 99.84% accuracy on the ISCXVPN2016 and ISCXTor2017 datasets.

Cheng et al. [30] introduce MATEC, a CNN-based framework with multi-head attention that enhances encrypted traffic classification while reducing computational complexity. Lotfollahi et al. [27] propose Deep Packet, integrating CNNs with Stacked Autoencoders (SAEs) to learn hierarchical traffic representations, achieving F1-scores of 0.98 for application classification and 0.94 for traffic characterization.

Several studies enhance CNN models by integrating hybrid architectures. Wang et al. [31] combine CNNs with SAEs to preserve essential session-level features, achieving an F1-score of 0.98. Soleymanpour et al. [32] introduce CSCNN, a cost-sensitive CNN that improves recall for underrepresented traffic classes. Xu et al. [33] propose ETC-Net, leveraging Siamese Convolutional Networks (SCNs) to improve classification with limited labeled data. Lin et al. [34] integrate CNNs with Bi-GRUs for improved spatial and sequential feature extraction, achieving 93.1% accuracy on the ISCX VPN-nonVPN dataset.

Advancements also focus on optimizing CNNs for real-world deployment. Hu et al. [35] introduce tCLD-Net, a CNN-LSTM model optimized with transfer learning, reducing training requirements while maintaining 86% accuracy. Zhu et al. [36] propose BCFNet, a CNN-transformer hybrid integrating ET-BERT and an Inception-inspired 1D-CNN, achieving 98.88% accuracy.

These CNN-driven approaches have significantly improved encrypted traffic classification. However, challenges such as computational efficiency, class imbalance, and generalizability persist, necessitating further optimization and hybrid methods.

Graph-Based Neural Networks

Graph-based neural networks (GNNs) have been explored for encrypted traffic classification, leveraging structured data representations to model traffic relationships and enhance classification accuracy [37, 38]. These models exploit network topology and traffic flow dependencies, addressing the limitations of conventional sequence-based classifiers.

Diao et al. [39] propose EC-GCN, a multi-scale Graph Convolutional Network (GCN) framework that converts traffic flows into structured graphs, achieving up to 20% accuracy improvement over state-of-the-art models on ISCX-Tor and other datasets. Hong et al. [40] introduce MalDiscovery, integrating GraphSAGE with multi-view feature fusion to construct a KNN-attributed graph for encrypted session classification, achieving 99.9% accuracy on the CTU-13 and MCFP datasets.

Wang et al. [41] present TGPrint, a GCN-based model using dynamic attack graphs to identify encrypted network threats, achieving 99% precision on CICIDS and HIKARI-2021 datasets. Han et al. [42] develop DE-GNN, a dual-embedding GNN framework that integrates packet- and flow-level learning for fine-grained classification, demonstrating superior performance on ISCX-VPN2016 and ISCX-Tor2016 datasets.

Zhang et al. [43] introduce CLE-TFE, a contrastive learning-enhanced framework that aligns packet- and flow-level features, reducing computational overhead by 14× while improving classification accuracy on ISCX VPN-nonVPN and ISCX Tor-nonTor datasets. Yang et al. [44] propose MTSecurity, a privacy-preserving encrypted malicious traffic classifier combining GNNs and Transformers to model encrypted traffic interactions, achieving 99.49% accuracy on MCFP and 99.46% on USTC-TFC datasets.

Graph-based neural networks provide a structured approach to encrypted traffic classification, capturing complex traffic relationships. However, their scalability and computational efficiency remain challenges, limiting real-time deployment. Future research should focus on optimizing these aspects to enhance practical applicability.

3.2.3 Generative adversarial networks (GANs) and transformers

Generative Adversarial Networks (GANs) and Transformers have emerged as effective methodologies in encrypted traffic classification (ETC), addressing challenges such as class imbalance, data augmentation, and automated feature extraction [45, 46]. These models enhance classification performance by leveraging the generative power of GANs and the sequential modeling capabilities of transformers.

Tang et al. [47] propose Markov-GAN, a framework that transforms encrypted traffic into structured grayscale images for CNN-based classification, employing GAN-based augmentation to improve class balance. The model achieves over 99% accuracy on ISCXVPN2016 and USTC-TFC2016 datasets. Wang et al. [48] introduce PacketCGAN, a conditional GAN that synthesizes minority-class traffic samples, improving CNN classifier performance to 99.51% accuracy, outperforming traditional balancing techniques.

Sanjalawe et al. [49] present a hybrid framework integrating Bidirectional GANs (BiGANs) for data augmentation and Vision Transformers (ViTs) for classification, achieving 99.59% accuracy on ISCX-Tor2016. Wang et al. [50] propose ByteSGAN, a semi-supervised GAN for encrypted traffic classification in SDN Edge Gateways, demonstrating superior performance with an F1-score of 99.06% on ISCX VPN-nonVPN and Crossmarket datasets.

Transformer-based approaches have also gained prominence. Zhao et al. [51] introduce MT-FlowFormer, a semi-supervised framework utilizing a FlowFormer encoder to model flow sequence dependencies, achieving 84.7% accuracy with only 10% labeled data. Lin et al. [52] develop ET-BERT, a transformer-based pre-training model that employs self-supervised learning tasks to improve VPN classification, achieving an F1-score of 98.9% on ISCX-VPN-Service and CSTNET-TLS 1.3 datasets.

Huang et al. [53] propose BSTFNet, a hybrid deep learning framework integrating BERT, BiGRU, and TextCNN to capture semantic, temporal, and spatial features, achieving 99.39% accuracy on USTC-TFC2016. Park et al. [54] introduce a DistilBERT-based multi-task learning framework that efficiently classifies encapsulation, category, and application levels, achieving 99.29%, 97.38%, and 96.89% accuracy, respectively.

GANs and transformers offer promising advancements in ETC, particularly in handling imbalanced datasets and capturing intricate traffic dependencies. However, challenges such as computational overhead and interpretability remain, necessitating further optimization for practical deployment.

3.3 Hybrid models

Hybrid models in encrypted traffic classification (ETC) integrate machine learning (ML) and deep learning (DL) to enhance accuracy, adaptability, and computational efficiency [55, 56]. These models offer robust solutions to evolving network security challenges by leveraging various analytical techniques.

Marim et al. [57] analyze ML models for darknet traffic classification, demonstrating that Decision Trees (DTs) achieve 99.89% accuracy on CIC-Darknet2020, outperforming deep learning classifiers in computational efficiency. Xu et al. [58] introduce ME-Box, a hybrid detection framework integrating ML-based classification with middlebox-driven evidence verification, reducing false positives while maintaining high detection accuracy.

Hu et al. [59] propose a hierarchical classifier integrating ML and DL to analyze user behavior across darknet environments, achieving 96.9% accuracy in darknet classification. Rust-Nguyen et al. [60] evaluate ML and DL classifiers against adversarial attacks, demonstrating the efficacy of SMOTE-based augmentation and GAN-generated samples in improving classifier robustness.

Automated architecture optimization has also been explored. Malekghaini et al. [61] present AutoML4ETC, a Neural Architecture Search (NAS)-based framework integrating reinforcement learning and Monte Carlo Tree Search for optimal model selection, improving classification accuracy on ISCX VPN-nonVPN and USTC-TFC2016 datasets. Elmaghraby et al. [62] investigate hybrid models that combine neural networks with ML classifiers, achieving 96.8% accuracy while improving computational efficiency.

Feature optimization techniques further enhance hybrid ETC frameworks. Luo et al. [63] propose BITization, a binary encoding mechanism that improves interpretability and classification accuracy by 23.7% on ISCX VPN-Service and USTC-TFC2016 datasets. Yan et al. [64] introduce HETC, a framework optimized for real-time classification using entropy-based feature selection, achieving 94% F-measure in encrypted traffic detection.

Recent advancements integrate meta-learning and incremental learning for enhanced adaptability. Zhao et al. [65] present METAROCKETC, an adaptive ETC framework utilizing Model-Agnostic Meta-Learning (MAML) for few-shot learning, outperforming deep learning models in evolving network environments. Li et al. [66] propose MISS, a multi-view sequence fusion model that improves long-term classification accuracy by 11.37% on PURE-TLS and USTC-TFC-2016 datasets.

Contrastive learning and multi-task fine-tuning have further optimized hybrid models. Wang et al. [67] introduce a spatial-temporal fusion framework combining 2D-CNNs for packet features and 1D-CNNs with LSTMs for sequential dependencies, achieving 99.9% accuracy for VPN classification. Wang et al. [68] propose a multi-task ETC framework utilizing Parameter-Efficient Fine-Tuning (PEFT) with Low-Rank Adaptation (LoRA), enhancing scalability across VPN recognition, malware detection, and intrusion classification.

Hybrid models significantly improve encrypted traffic classification by combining ML, DL, and meta-learning techniques, enhancing robustness and efficiency. However, challenges such as computational overhead and interpretability remain, necessitating further optimization for real-world deployment.

3.4 General discussion

The extensive review of existing encrypted traffic classification approaches reveals a clear progression from traditional machine learning models toward deep learning, hybrid models, and explainable AI-driven frameworks. Early works, such as Zaki et al. [11] and Yao et al. [13], achieved promising results using handcrafted features and ensemble learning. Still, their adaptability to evolving encryption protocols remains limited. Deep learning-based models, including autoencoders and CNN architectures [16, 28], improved feature extraction and classification performance, yet often suffer from computational overhead and interpretability challenges.

Recent advancements, such as EC-GCN by Diao et al. [39] and MT-Security by Yang et al. [44], demonstrate the effectiveness of graph-based and transformer-based methods for capturing complex traffic relationships and enhancing classification robustness. However, these approaches tend to introduce significant computational complexity, potentially limiting real-time deployment feasibility.

Compared to these prior studies, the proposed Hybrid Attention + LightGBM model offers a balanced approach by integrating attention-based feature enhancement with an efficient tree-based classifier. This design provides strong

Table 1 Comparative Summary of Existing Works in Encrypted Traffic Classification

Paper	Methodology/Focus	Strengths	Weaknesses
He et al. (2020) [28]	CNN with Grayscale Image Transformation	High classification accuracy	Image generation overhead
Yao et al. (2020) [13]	GMM + HMM for protocol identification	Robust encrypted protocol identification	Limited adaptability to novel traffic
Shi Dong (2021) [12]	Cost-Sensitive SVM + Active Learning	Enhanced recall for minority classes	Computational cost of active learning
Aceto et al. (2021) [16]	DISTILLER (Multimodal DL Classifier)	Multimodal feature extraction; improved robustness	Complex training pipeline
Zaki et al. (2022) [11]	Classifier Chain with Random Forests	High multi-label classification accuracy; scalable	Limited to handcrafted features
Tang et al. (2022) [47]	Markov-GAN for Data Augmentation	Effective minority class balancing	GAN training instability
Lv et al. (2023) [15]	AAE-DSVDD (Autoencoder + One-Class)	Effective anomaly detection; unsupervised learning	Risk of hypersphere collapse
Diao et al. (2023) [39]	EC-GCN (Graph Convolution for Traffic)	Captures traffic structure dependencies	Scalability issues for large graphs
Wang et al. (2024) [68]	PEFT for Multi-Task ETC	Improved scalability with fewer parameters	Slight drop in performance on rare classes
Yang et al. (2024) [44]	GNN + Transformer (MT-Security)	High accuracy and privacy preservation	High computational overhead
Our Work (Hybrid Attention + LightGBM)	Attention-based feature fusion + XAI-guided LightGBM	High accuracy, interpretability, balanced class performance, real-world feasibility	Requires optimization for real-time large-scale deployment

classification performance while maintaining computational efficiency and model interpretability, as validated through SHAP and LIME analyses. Additionally, targeted optimization strategies, such as class-specific threshold adjustments and selective feature pruning, further enhance the model's performance, particularly in addressing class imbalance and improving recall for underrepresented traffic types like I2P and Freenet.

Nevertheless, consistent with broader trends observed in the literature, challenges remain regarding feature overlap, scalability to huge network environments, and resilience against adversarial evasion techniques. Future improvements could benefit from further integrating advanced meta-learning strategies, lightweight architecture designs, and continual learning approaches to enhance adaptability and deployment feasibility in dynamic network conditions.

This study builds upon and extends state-of-the-art methodologies, addressing existing limitations while contributing new insights into scalable, interpretable, and high-performance encrypted traffic classification.

3.5 Synthesis

Encrypted traffic classification (ETC) remains a challenging task in network security. Computational efficiency and resource constraints significantly impact the real-world deployment of classification models. This section highlights key limitations affecting real-time performance, scalability, and adaptability.

Significant challenges include high computational overhead, extensive preprocessing requirements, and the dependency on diverse, high-quality datasets. Additionally, ensuring model scalability, continuous adaptation to evolving traffic patterns, and real-time feasibility remain critical concerns. Addressing these limitations is essential for advancing ETC methodologies toward more efficient and deployable solutions. The following points outline these limitations in order of importance:

1. Real-Time Application Feasibility -[15, 40, 50, 59, 69–73].
2. High Computational Complexity -[15, 24, 27, 29, 35, 41, 43, 44, 48, 53, 74].

3. Scalability Concerns Due to Advanced Features - [51, 75].
4. Adaptation Challenges in Evolving Network Conditions - [60, 65].
5. Impact of Data Quality and Diversity on Model Effectiveness - [16, 21, 30, 40, 49, 57, 61, 64, 68, 76].
6. Resource-Intensive Preprocessing and Model Architecture - [21, 54, 77].
7. Challenges in Data Collection and Representation - [11, 28, 32, 52].
8. Complex Model Integration and Scalability Issues - [19, 22, 33, 34, 41, 62, 78, 79].
9. Need for Ongoing Model Updates - [12, 19, 23, 31, 47, 66, 80].
10. Practical Deployment Issues in Dynamic Networks - [20, 39, 47].

Despite advancements in deep learning and hybrid models, issues such as adversarial robustness, explainability, and real-time deployment remain at the forefront of research. Addressing these challenges requires scalable and efficient classification techniques that minimize computational overhead while ensuring high performance and adaptability. Future work should optimize feature selection, improve model interpretability, and enhance computational efficiency to enable effective deployment in real-world network environments.

4 Proposed model

This section presents the design, implementation, and evaluation strategies for an interpretable and high-performance encrypted traffic classification model.

The proposed approach integrates Explainable Artificial Intelligence (XAI) within a Hybrid Attention + LightGBM architecture. Multi-head attention mechanisms enhance feature extraction, while LightGBM ensures efficient and scalable classification.

The classification follows a hierarchical two-layer process:

Layer 1: Binary Classification: Distinguishes encrypted from non-encrypted traffic.

Layer 2: Multi-Class Classification: Categorizes encrypted traffic into VPN, Tor, I2P, ZeroNet, and Freenet.

Using SHAP and LIME for feature importance analysis enhances model interpretability. SHAP optimizes feature selection and pruning, improving classification efficiency, while LIME provides insights into misclassified instances, refining the model for real-world deployment.

4.1 Architecture design

The proposed model architecture, illustrated in Figure 2, follows a multi-stage pipeline for efficient classification of encrypted and non-encrypted traffic, with further categorization into sub-classes. The model integrates feature extraction, preprocessing, attention-based feature fusion, and classification while incorporating explainability techniques to enhance interpretability.

The classification workflow consists of the following key components:

1. **Feature Extraction via NTLFlowLyzer:** Network traffic features are extracted using NTLFlowLyzer, generating over 400+ handcrafted features at both packet and flow levels [81]. For this project, we developed a new version of NTLFlowLyzer-V0.3.0 including the entropy-based statistical features extracted from encrypted traffic using the entropy functions [82]. This dataset serves as the foundation for preprocessing and classification.
2. **Data Preprocessing:** Raw features undergo *missing value handling, feature scaling, and data transformation* to improve model consistency and performance.
3. **Feature Selection and Attention Mechanism: Mutual Information (MI)-based feature selection** reduces computational complexity while retaining relevant features. A Multi-Head Attention (MHA) mechanism enhances feature interactions, generating an optimized feature representation.
4. **Classification Framework:** The final classification uses LightGBM, ensuring high efficiency and scalability. Attention-based feature fusion enhances classification accuracy.
5. **Explainability and Evaluation: SHAP and LIME** provide feature importance analysis and interpretability, while classification performance is evaluated using accuracy, precision, recall, and F1-score.

The subsequent subsections provide detailed insights into each stage, starting with the CSV generation process using NTLFlowLyzer [81], outlining how raw network traffic is processed to extract meaningful features for classification.

4.1.1 CSV generation using NTLFlowLyzer

The first step in the proposed architecture involves extracting relevant features from raw network traffic data using NTLFlowLyzer -V 0.3.0 [82]. This network traffic analyzer processes packet capture (PCAP) files into structured flow-based feature representations. These features capture critical network characteristics, enabling effective encrypted and non-encrypted traffic classification.

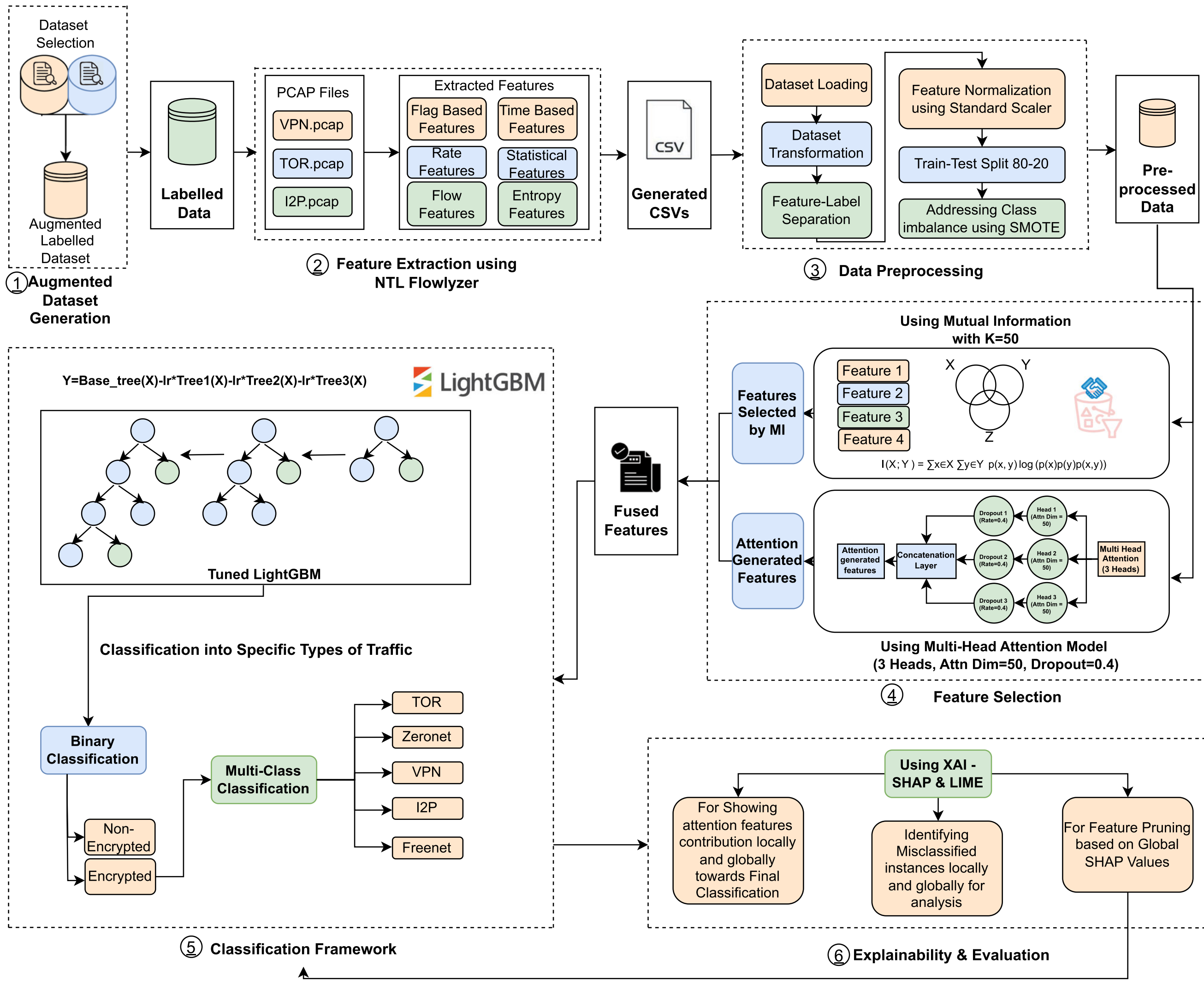

Fig. 2 Architecture of the proposed Multi-Head Attention model with feature selection using Mutual Information (MI) and attention-based feature fusion, integrated with LightGBM for encrypted traffic classification.

Feature Extraction Workflow Network traffic is collected in PCAP files, containing packet-level data, including protocol headers, payloads, timestamps, and flow information. NTLFlowLyzer processes these files to extract a diverse set of statistical, temporal, and protocol-based features [81]. A flow is defined as a bidirectional exchange of packets between two endpoints, aggregated based on IP addresses, ports, and protocols. Extracted features capture essential traffic attributes such as packet counts, inter-packet delays, and entropy measures, supporting robust classification.

Feature Categorization Extracted features are systematically grouped to enhance interpretability and optimize classification:

1. Basic Flow Statistics: Includes duration, packet counts, and bytes transferred, providing fundamental insights into network traffic.
2. Statistical Measures: Captures traffic variability through mean, variance, skewness, and kurtosis to detect anomalies and trends.
3. Protocol-Specific Features: Extracts TCP flag statistics, handshake metadata, and initial congestion window sizes for protocol analysis.
4. Advanced Metrics: Includes entropy analysis, n-gram entropy, and mutual information to evaluate encryption complexity.
5. Directional Analysis: Computes features separately for forward, backward, and aggregated flows, offering a comprehensive view of traffic behavior.

The dataset ensures high-quality input for subsequent preprocessing and classification by structuring extracted features into meaningful categories. The following subsection, Data Preprocessing, details the steps taken to refine and optimize these features for classification.

4.1.2 Data preprocessing

Data preprocessing ensures a structured and balanced dataset for classification by refining raw network traffic data. The dataset, extracted using NTLFlowLyzer, retains only relevant columns (starting from the **7th index**) to enhance computational efficiency [81]. Categorical features (excluding labels) are transformed using string length conversion, while missing values (NaN, inf, -inf) are replaced with the median of their respective features to maintain consistency. The dataset is structured by separating features (X) from labels (Y), with the `label` column serving as the ground truth, and categorical labels are encoded using LabelEncoder for compatibility. To address scale disparities, StandardScaler normalizes feature values to a mean of zero and a standard deviation of one, ensuring improved convergence. The dataset is then split into training and testing sets using an 80-20 ratio with a fixed random state for reproducibility. To mitigate class imbalance, the Synthetic Minority Oversampling Technique (SMOTE) generates synthetic samples for underrepresented classes, improving generalization and preventing biased predictions. These preprocessing steps ensure an optimized dataset for classification, enhancing model performance and interpretability.

4.1.3 Feature selection

Feature selection is crucial for reducing dimensionality, eliminating irrelevant features, and improving model efficiency. The proposed approach integrates Mutual Information (MI) and Multi-Head Attention (MHA) to retain the most informative attributes for encrypted traffic classification. MI quantifies feature dependencies with the target variable, capturing linear and non-linear relationships to enhance interpretability. The top $k = 50$ features were selected using `SelectKBest` with the `mutual_info_classif` scoring function, applied to the resampled training dataset to prevent bias due to class imbalance. MHA dynamically assigns attention weights to refine feature representation further, capturing intricate dependencies that improve classification accuracy. Each attention head independently learns feature interactions, and their outputs are concatenated to form an enriched representation, with a dropout rate of 0.4 applied to mitigate overfitting. The final feature set fuses MI-selected attributes with attention-enhanced representations, improving classification accuracy, regularization, and generalization. This integration strengthens the model's ability to distinguish encrypted traffic types effectively. The following section details the classification framework, incorporating these optimized features into the learning process.

4.1.4 Classification framework and attention mechanism

The proposed model employs a two-layer classification framework, where Layer 1 distinguishes between encrypted and non-encrypted traffic, and Layer 2 classifies encrypted traffic into specific categories. An attention mechanism enhances feature representations, improving classification accuracy and interpretability.

Tuned LightGBM Model Initialization and Training A tuned LightGBM classifier is optimized via Randomized Search CV to balance accuracy and generalization. Key hyperparameters include colsample _bytree (0.9249), learning_rate (0.0691), num _leaves (47), n_ estimators (383), and regularization terms (reg _alpha = 0.2184, reg_lambda = 0.4165). The model is trained on `X_train_fused` and `y_train_res` to ensure class balance, leveraging LightGBM's efficiency in processing high-dimensional data.

Two-Layer Classification Process Layer 1 includes Binary Classification which Separates encrypted from non-encrypted traffic using attention-fused features. LightGBM evaluates performance using precision, recall, and F1-score, prioritizing recall to minimize false negatives.

The Layer 2 for Multi-Class Classification, categorizes encrypted traffic into Tor, VPN, I2P, ZeroNet, and Freenet. A fine-tuned LightGBM model applies class-specific weighting and SMOTE to address the class imbalance, optimizing predictions via macro-F1 scoring and confusion matrix analysis.

Attention Mechanism and Feature Fusion The Multi-Head Attention (MHA) mechanism refines feature representations by dynamically prioritizing critical attributes, leading to several key benefits. First, it provides enhanced feature representation by emphasizing influential features, and improving model interpretability. Additionally, MHA contributes to improved classification accuracy by capturing intricate feature dependencies, ensuring precise predictions. Finally, it enhances robustness to class imbalance through adaptive feature weighting, strengthening model generalization, and providing reliable classification across diverse encrypted traffic categories.

Model Validation and Interpretability Feature importance is validated using SHAP and LIME, confirming the contribution of attention-fused features to classification accuracy. Integrating an optimized LightGBM model with attention-based feature refinement ensures high-performance encrypted traffic classification.

The following subsection details model evaluation and interpretability metrics.

4.1.5 Explainability and evaluation

Ensuring model transparency is critical for encrypted traffic classification, particularly in security applications. The proposed model integrates Explainable AI (XAI) techniques, precisely SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations), to validate feature importance and improve interpretability. These methods confirm the significance of attention-based feature fusion in refining classification accuracy.

Explainability in Layer 1: Binary Classification SHAP and LIME are utilized in Layer 1 to perform the following analyses:

1. SHAP for Feature Importance: SHAP assesses global and local feature contributions in differentiating encrypted from non-encrypted traffic. Using `Tree Explainer`, SHAP values are computed for the LightGBM classifier, revealing key influential features through summary and force plots.
2. LIME for Local Interpretability: LIME supplements SHAP by providing instance-level explanations, detailing feature contributions to classification outcomes. A `LimeTabularExplainer` analyzes misclassifications, enhancing sample-level transparency.
3. Feature Pruning with SHAP: SHAP-based pruning removes low-importance features, optimizing model efficiency while preserving accuracy. The LightGBM model is retrained on the pruned dataset, with SHAP and LIME analyses confirming the retained features' predictive relevance.

Explainability in Layer 2: Multi-Class Classification SHAP and LIME are utilized in Layer 2 to perform the following analyses:

1. SHAP and LIME for Feature Importance: Class-specific analysis uses SHAP and LIME values, computed separately for each encrypted traffic category (Tor, VPN, I2P, Freenet, and Zeronet), identifying class-specific influential features via summary plots.
2. Feature Pruning and Refinement: SHAP-based ranking retains the top 75% most informative features, reducing computational overhead while maintaining classification accuracy. SHAP and LIME confirm the post-pruning model's robustness.
3. Misclassification Analysis and Model Refinements: SHAP and LIME are leveraged to diagnose misclassifications, providing insights at both global and local levels. SHAP summary plots highlight features frequently associated with errors, offering a comprehensive view of misclassification trends. Meanwhile, SHAP force plots and LIME visualizations facilitate local analysis by revealing feature interactions responsible for specific misclassifications. Targeted refinements such as class-specific threshold tuning and data augmentation are applied to enhance model robustness and minimize errors. SHAP ensures global feature attribution, while LIME provides instance-level interpretability, enabling transparency across both dataset-wide patterns and individual sample predictions. Post-classification insights further validate the effectiveness of the attention mechanism in feature selection, demonstrating the proposed approach's ability to maintain an optimal balance between accuracy, efficiency, and interpretability.

5 Experiments and results

This section presents the experimental setup, dataset selection, model training, and evaluation results. The proposed model is assessed on the augmented datasets, focusing on classification performance, interpretability, and computational efficiency. The evaluation follows a structured methodology, ensuring reproducibility and robustness in real-world encrypted traffic scenarios.

5.1 Augmented dataset creation

High-quality datasets are essential for training robust encrypted traffic classification models. This subsection details dataset selection, augmentation, and preprocessing to ensure a well-balanced dataset representative of real-world encrypted traffic patterns. Publicly available datasets were evaluated based on encryption diversity, labeling consistency, and real-world applicability. CSV files were generated using NTLFlowlyzer to structure data for analysis and model training [81].

The growing prevalence of encrypted traffic demands datasets encompassing diverse encryption protocols, attack scenarios, and network conditions. A structured evaluation process was conducted to identify datasets that optimize classification performance across machine learning (ML), deep learning (DL), and hybrid models to achieve comprehensive coverage.

5.1.1 Augmented dataset creation (BCCC-Darknet-2025)

Based on the selection criteria (Table 2), CIC-Darknet 2020 [83] and Darknet-Dataset-2020 [84] were chosen for their extensive representation of encrypted traffic types, including Tor, I2P, Freenet, and ZeroNet. These datasets provide multi-class labeling, capture darknet-specific activities, and preserve temporal dependencies, covering entropy, time-based, and protocol-specific features. Below is a brief description of the selected datasets:

Table 2 Shortlisted Dataset Analysis Table

Dataset	Labelled		Encryption Service						Capture		Metadata		
	Binary	**Multi**	**VPN**	**TOR**	**I2P**	**Zeronet**	**Freenet**	**Others**	**Complete**	**Partial**	**Docs**	**PCAP**	**CSV**
ISCX Tor/Non-Tor (2016)	Y	N	N	Y	N	N	N	N	Y	N	Y	Y	Y
ISCX VPN/Non-VPN (2016)	Y	N	Y	N	N	N	N	N	Y	N	Y	Y	Y
SJTU - AN21 (2021)	N	Y	N	Y	Y	N	N	Y	Y	N	Y	N	Y
Darknet-dataset (2020)	N	Y	N	Y	Y	Y	Y	N	Y	N	Y	Y	Y
CIC-Darknet (2020)	N	Y	Y	Y	N	N	N	N	Y	N	Y	Y	Y

Table 3 Performance Metrics of the Tuned Hybrid Attention + LightGBM Model for Binary Classification

Class	Precision	Recall	F1-score	Support
Encrypted	0.97	0.99	0.98	1370
Non-Encrypted	0.99	0.99	0.99	3738
Accuracy		0.99		
Macro Avg		0.98		
Weighted Avg		0.99		

1. **CIC-Darknet2020:** A comprehensive dataset covering multiple encryption protocols, including Tor and VPN, supporting network forensics and cybersecurity research [83].
2. **Darknet-Dataset-2020:** Focuses on behavioral aspects of encrypted traffic, improving detection accuracy by identifying unique communication patterns [84].

The integration of these datasets strengthens model generalization, particularly in detecting encrypted traffic anomalies. Their structured selection ensures a dataset that effectively supports encrypted traffic classification under evolving network conditions.

NTLFlowlyzer has been applied to these two datasets to standardize feature extraction, ensuring seamless dataset integration into the classification pipeline. The resulting augmented dataset (BCCC-Darknet-2025) enhances model generalizability, improving classification accuracy in dynamic cybersecurity environments [81].

5.2 Results

This section evaluates the optimization strategies and their impact on model performance for binary classification (Layer 1) and multi-class classification (Layer 2). It presents the results of SHAP and LIME analyses, feature pruning strategies, and the performance evaluation of the optimized models, demonstrating their effectiveness in improving classification accuracy and interpretability.

5.2.1 Layer 1 - binary classification

The first layer of the model performs binary classification, distinguishing between encrypted and non-encrypted traffic. This stage ensures accurate filtering and preprocessing before multi-class categorization. The Tuned Hybrid Attention + LightGBM model was extensively evaluated to ensure high accuracy and reliability.

```
********************************
Cross-Validation F1 Scores (Tuned LightGBM Model): [0.98449351 0.98516769 0.98550391 0.98381863 0.98718796]
Mean F1 Score (Tuned LightGBM Model): 0.98523434106082
********************************
```

Fig. 3 Cross-validation F1 scores of the Tuned LightGBM model for binary classification.

Tuned LightGBM Model Evaluation The evaluation of the Tuned Hybrid Attention + LightGBM model demonstrated high performance in binary classification. As shown in Table 3, the model achieved an accuracy of 99% and an F1-score of 0.98 for Encrypted traffic and 0.99 for Non-Encrypted traffic. Cross-validation analysis further confirmed model stability, with a mean F1-score of 0.9852 over five folds (Fig. 3), reinforcing its robustness and generalization capability. These results validate the model's effectiveness in distinguishing encrypted traffic, forming a reliable foundation for the next classification stage.

SHAP-Based Feature Importance and Pruning

SHAP analysis was conducted to optimize model efficiency while preserving classification accuracy. The most influential features identified included `x239`, `x87`, `x91`, and `attention_feature_22`, demonstrating the hybrid model's ability to integrate both original and attention-enhanced features (Fig. 4). Attention-derived features, such as `attention_feature_17` and `attention_fea--ture_41`, provided complementary insights, improving classification of ambiguous instances. Additionally, interactions between features like `attention_feature_28` and `x92` highlighted the synergy between handcrafted and learned features, reinforcing their combined role in refining classification decisions.

SHAP-based Pruning Strategy and Retraining

A pruning strategy was applied using SHAP feature importance values to enhance model efficiency and reduce computational complexity. Post-classification SHAP analysis identified minimally contributing features removed based on a quantile-based threshold, retaining only the top 75% most influential features. The model was then retrained on the reduced feature set, and performance metrics, including F1-score and accuracy, were compared against the baseline model to evaluate the impact of pruning on classification performance.

The results (Table 4 and Fig. 5) demonstrated that the pruned model retained its predictive capability, validating the efficacy of SHAP-driven pruning in maintaining classification accuracy while reducing computational complexity.

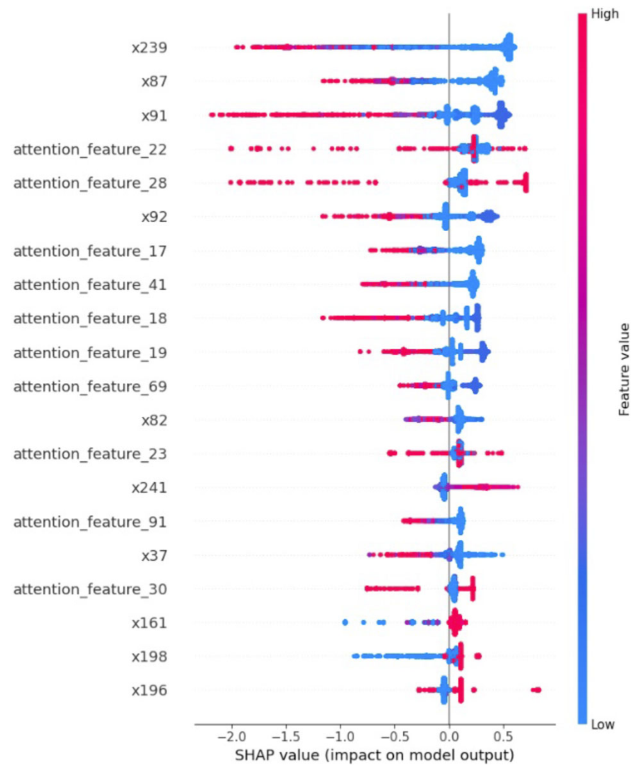

Fig. 4 SHAP summary plot illustrating the impact of various features, including attention-based features, on the model's output for Layer 1.

Table 4 Performance Metrics of the Pruned Tuned Hybrid Attention + LightGBM Model after SHAP-Based Feature Pruning

Class	Precision	Recall	F1-score	Support
Encrypted	0.97	0.99	0.98	1370
Non-Encrypted	0.99	0.99	0.99	3738
Accuracy		0.99		
Macro Avg		0.98		
Weighted Avg		0.99		

5.2.2 Layer 2 - multi-class classification

The second layer of the model performs multi-class classification, categorizing encrypted traffic into VPN, Tor, I2P, ZeroNet, and Freenet. Unlike binary classification, this task involves distinguishing between encrypted services with overlapping patterns, increasing classification complexity.

Tuned LightGBM Model Evaluation

The Tuned Hybrid Attention + LightGBM model demonstrated strong multi-class classification performance, achieving an overall accuracy of 93% and a macro-average F1-score of 0.90. A detailed breakdown (Table 5) highlights variations

```
Cross-Validation F1 Scores (Pruned Model): [0.98398785 0.98668462 0.98668406 0.98483003 0.98701939]
Mean F1 Score (Pruned Model): 0.9858411890250196
```

Fig. 5 Testing metrics and cross-validation results for the pruned LightGBM model, demonstrating performance after SHAP-based feature pruning.

Table 5 Performance Metrics of the Tuned Hybrid Attention + LightGBM Model for Multi-Class Classification

Class	Precision	Recall	F1-score	Support
FREENET	0.78	0.76	0.77	33
I2P	0.86	0.93	0.89	403
TOR	0.92	0.85	0.89	264
VPN	0.99	1.00	0.99	430
ZERONET	0.97	0.94	0.95	455
Accuracy		0.93 (1585)		
Macro Avg		0.90 (1585)		
Weighted Avg		0.93 (1585)		

in precision, recall, and F1-score across different encrypted services, reflecting class-specific challenges in distinguishing traffic types. Cross-validation analysis further confirmed model stability, with a mean F1-score of 0.9450 over five folds (Fig. 6), reinforcing its robustness and generalization capability.

SHAP and LIME-Based Feature Importance Analysis

SHAP and LIME analyses were conducted to enhance interpretability and understand feature contributions in multi-class classification. These techniques provided insights into global and class-specific feature influences on model predictions.

1. SHAP Analysis - SHAP analysis identified key features influencing classification across all traffic categories (Fig. 7). Features such as `x91`, `x92`, and `x199` consistently ranked among the top contributors, demonstrating their critical role in distinguishing encrypted traffic. Attention-enhanced features, particularly `attention_feature_19` and `attention_feature_25`, significantly impacted model decisions, validating the effectiveness of hybrid feature extraction. Class-specific SHAP insights (Figs. 8-12) further highlighted the role of attention-based features in distinguishing encrypted traffic types. For ZERONET, `attention_feature_25` played a crucial role, reinforcing the importance of learned representations. In the case of I2P, features such as `x65` and `x198` were instrumental in improving recall, particularly in differentiating I2P from TOR and ZERONET. These findings validate the hybrid model's feature extraction process and provide critical insights into how specific features influence classification outcomes.
2. LIME-Based Interpretability - LIME analysis was conducted to provide local explanations for individual predictions, highlighting the importance of both original and attention-enhanced features in classification decisions (Fig. 13). The results demonstrated that I2P classification relied heavily on key features such as `x198`, `attention_feature_25`, and `x65`, validating the role of hybrid feature integration in improving accuracy. LIME visualizations also provided insights into the model's confidence levels, confirming that high- confidence classifications aligned with expected outcomes. This reinforces the model's reliability in distinguishing encrypted traffic categories while maintaining high transparency and interpretability, ensuring trust in its decision-making process.

SHAP-based Pruning Strategy and Retraining To enhance model efficiency while preserving performance, SHAP-based pruning retained approximately 75% of the most influential features based on SHAP mean values. Features such as `x91`, `x92`, and `attention_ feature_19` were preserved due to their consistently high importance scores, ensuring critical classification information was maintained. This strategy effectively reduced feature dimensionality, optimizing computational efficiency without compromising accuracy.

The pruned model was retrained and evaluated against the baseline, achieving an overall accuracy of 93% and a macro-average F1-score of 0.90, confirming that feature reduction did not negatively impact classification performance (Table 6 and Fig. 14).

Class-specific performance remained strong across encrypted traffic categories. The classification of FREENET achieved a precision of 86%, with a recall of 73% and an

```
********************************
Cross-Validation F1 Scores (Tuned LightGBM Model): [0.9459111  0.9365302  0.95205187 0.95071753 0.93953519]
Mean F1 Score (Tuned LightGBM Model): 0.9449491781314764
********************************
```

Fig. 6 Cross-validation F1 scores of the Tuned LightGBM model for multi-class classification.

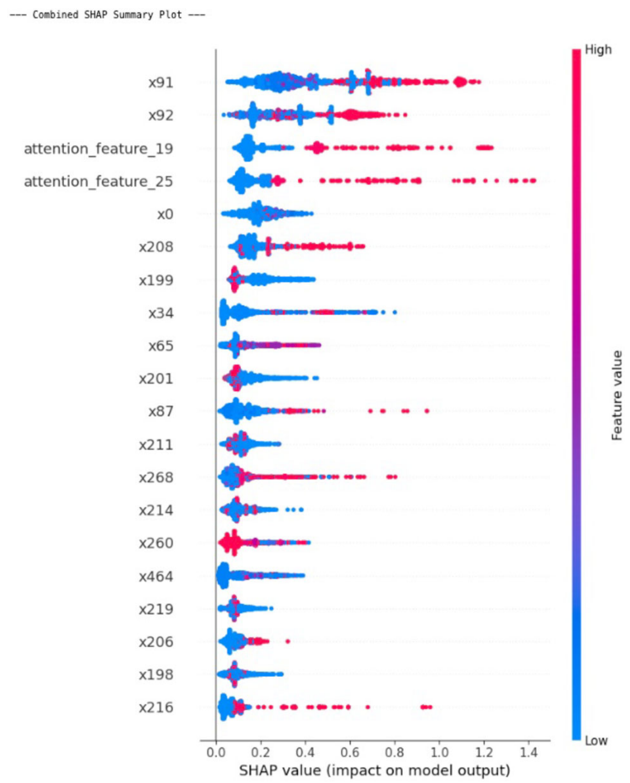

Fig. 7 Combined SHAP summary plot for multi-class classification, illustrating the impact of various features, including attention-based features, on the model's output for all Classes.

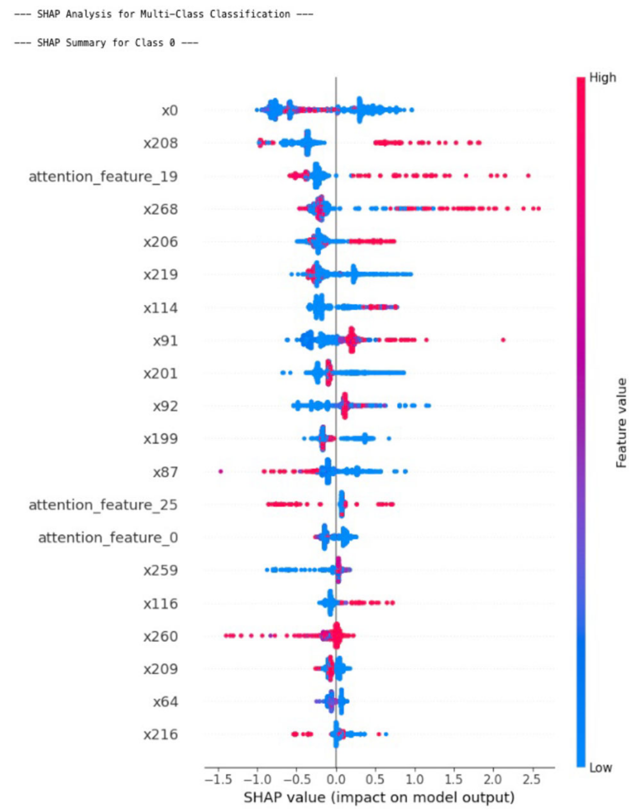

Fig. 8 SHAP summary plot for multi-class classification, illustrating the impact of various features on the model's output for Class 0.

F1-score of 79%, reflecting balanced performance despite its lower recall. I2P demonstrated high recall (92%) and precision (86%), resulting in an F1-score of 89%, indicating the model's effectiveness in identifying I2P traffic. Similarly, TOR attained a precision of 92% and recall of 86%, achieving an F1-score of 89%, showing strong performance. The classification of VPN exhibited near-perfect results with a precision of 99%, recall of 100%, and an F1-score of 99%, highlighting the model's accuracy in distinguishing VPN traffic. Finally, ZERONET achieved a precision of 96%, recall of 93%, and an F1-score of 95%, demonstrating excellent performance across all classes.

These results confirm that SHAP-driven feature selection optimizes model efficiency while maintaining high classification performance.

5.2.3 Evaluation using imbalance-sensitive metrics

To provide a more nuanced assessment of the model's ability to handle class imbalance, additional metrics including the Matthews Correlation Coefficient (MCC), the Geometric Mean (G-Mean), and the Precision-Recall Area Under Curve (PR-AUC) were computed for both binary and multiclass classification tasks. For the binary classification stage, the model achieved an MCC of 0.9699 and a G-Mean of 0.9886, indicating robust predictive performance with minimal bias toward either class. For multiclass encrypted traffic classification, the model achieved a PR-AUC of 0.9652, an MCC of 0.9055, and a G-Mean of 0.8877, further highlighting its effectiveness across all traffic types, including minority classes such as I2P and Freenet. The PR-AUC for binary classification was not computed due to the unavailability of stored prediction probabilities. The additional metrics substantiate the model's resilience against class imbalance and reinforce its practical applicability.

6 Analysis & discussion

This section comprehensively analyzes the proposed model's performance in encrypted traffic classification. The discussion is structured into two key aspects: the impact of optimization strategies on model refinement and a comparative evaluation against baseline classifiers.

The first part, Comparative Analysis of Optimization Approaches, examines the effects of weight tuning, threshold adjustments, and feature augmentation. These refinements

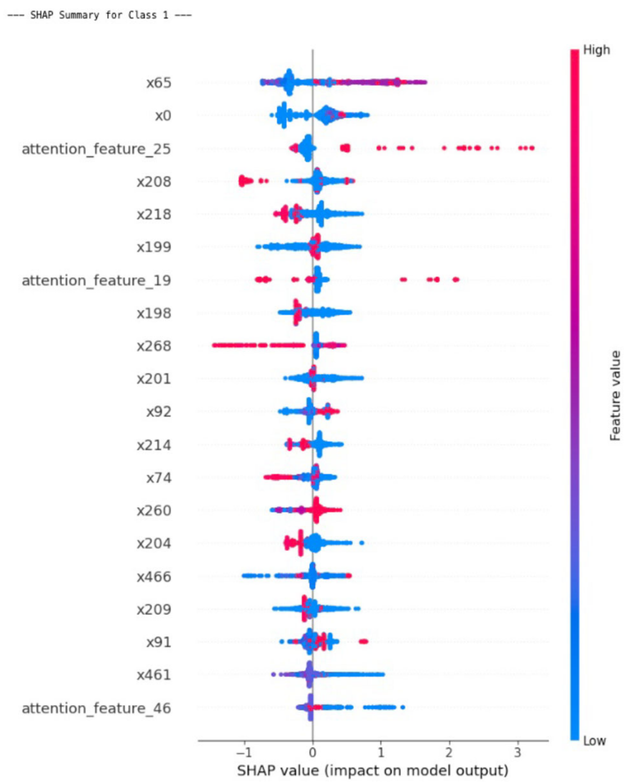

Fig. 9 SHAP summary plot for multi-class classification, illustrating the impact of various features on the model's output for Class 1.

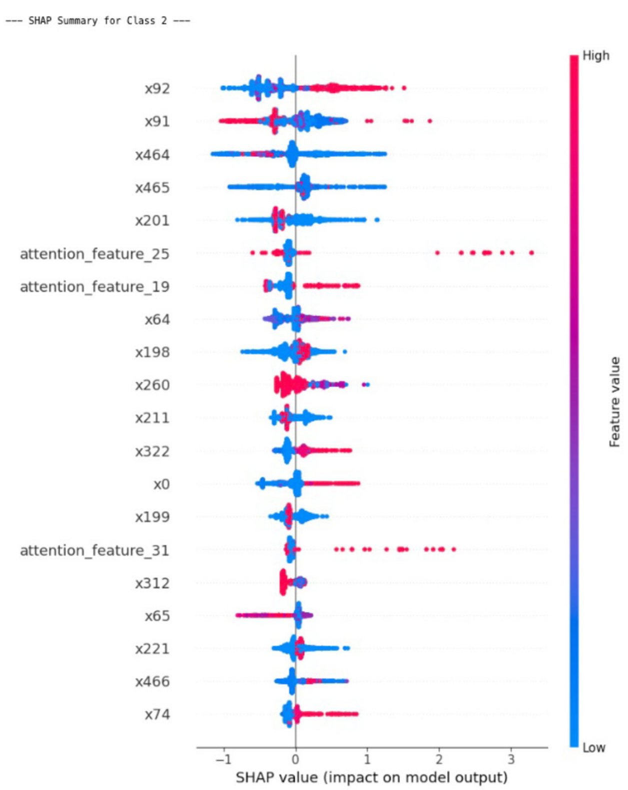

Fig. 10 SHAP summary plot for multi-class classification, illustrating the impact of various features on the model's output for Class 2.

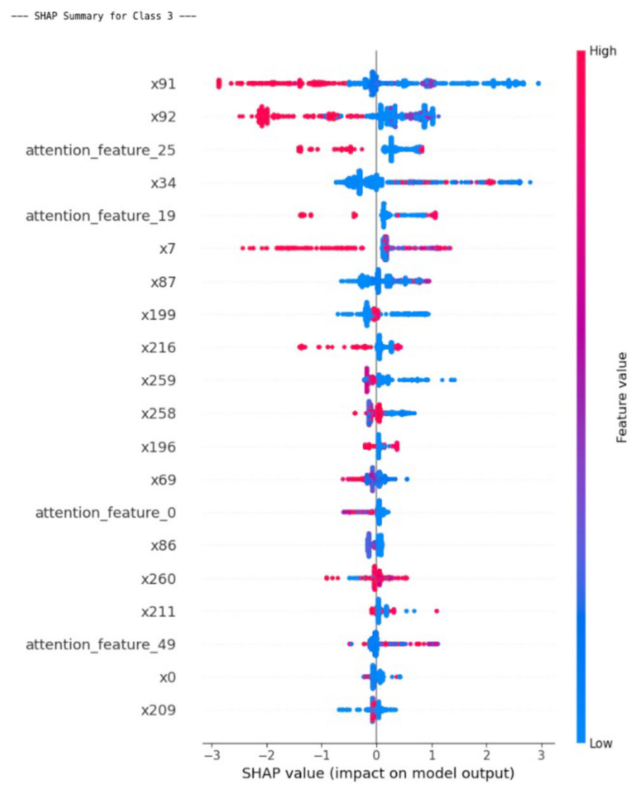

Fig. 11 SHAP summary plot for multi-class classification, illustrating the impact of various features on the model's output for Class 3.

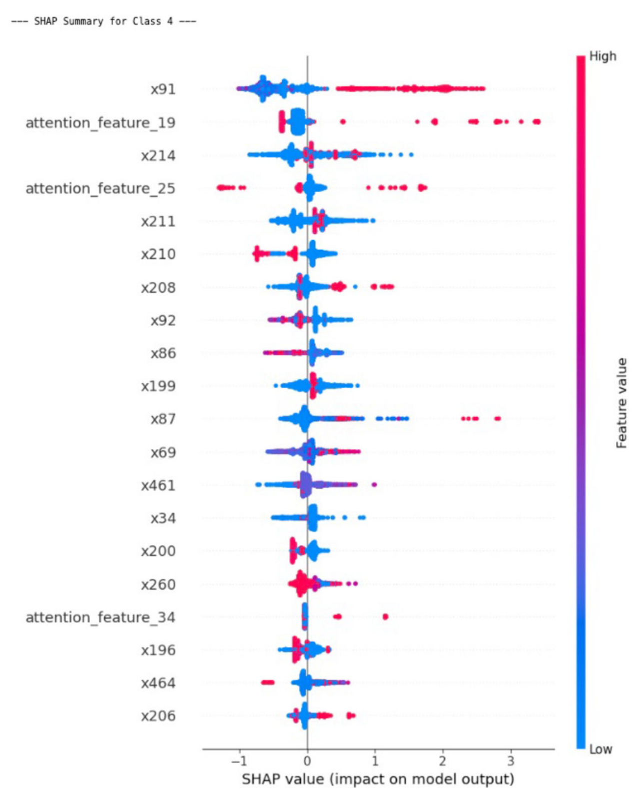

Fig. 12 SHAP summary plot for multi-class classification, illustrating the impact of various features on the model's output for Class 4.

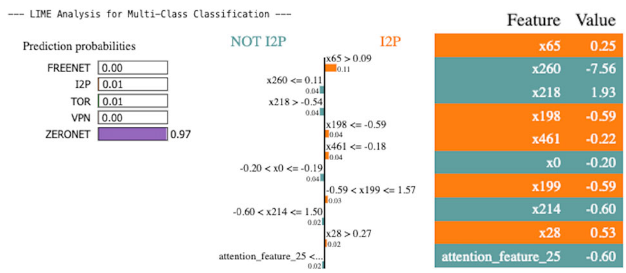

Fig. 13 LIME Analysis for Multi-Class Classification.

Table 6 Performance Metrics for the Pruned LightGBM Model, Demonstrating Performance After SHAP-Based Feature Pruning for Multi-Class Classification

Class	Precision	Recall	F1-score	Support
FREENET	0.86	0.73	0.79	33
I2P	0.86	0.92	0.89	403
TOR	0.92	0.86	0.89	264
VPN	0.99	1.00	0.99	430
ZERONET	0.96	0.93	0.95	455
Accuracy		0.93 (1585)		
Macro Avg		0.90 (1585)		
Weighted Avg		0.93 (1585)		

address the class imbalance, enhance recall for minority classes, and optimize the precision-recall trade-off.

The second part, Comparative Model Performance, benchmarks the hybrid LightGBM-based classifier against conventional machine learning and deep learning models. This analysis covers overall performance trends, the effectiveness of hybrid gradient boosting, class-wise recall comparisons, and confusion matrix insights. Additionally, the influence of architectural choices on classification accuracy is discussed, highlighting key takeaways and model efficiency insights.

6.1 Comparative analysis of three optimization approaches based on SHAP and LIME insights

Three distinct approaches-Weight Tuning, Threshold Tuning, and SMOTE Augmentation-are applied individually to evaluate the impact of different optimization techniques. This analysis isolates their effects on class balance, classification accuracy, and model interpretability.

1. **Visual Analysis of Feature Distributions** A visual analysis of feature distributions is conducted using Kernel Density Estimation (KDE) plots and pairwise scatterplots of key features (`x91`, `x198`, `x65`) to detect class overlap and potential sources of misclassification. These insights inform subsequent refinements by highlighting feature-space inconsistencies.
2. **Weight Tuning and Threshold Adjustments** To mitigate the disproportionate influence of VPN traffic, threshold adjustments are applied, reducing its weight in the classification process. The effect is evaluated through class-wise precision, recall, and F1-score changes, ensuring VPN does not overshadow minority classes such as I2P and TOR.
3. **SMOTE Augmentation for Minority Classes** SMOTE augmentation is applied to the FREENET class to enhance recall by generating synthetic samples. Class weighting is incorporated into the LightGBM loss function to complement SMOTE, ensuring proportional influence during training. Performance improvements are assessed based on reductions in false negatives.
4. **Class-Specific Threshold Tuning for I2P and ZERONET** Given their significant feature overlap, thresholds for I2P and ZERONET are dynamically adjusted to balance precision and recall. SHAP force plots guide threshold tuning by identifying key features influencing classification decisions and preserving distinct class boundaries while maintaining model generalization.

   Each optimization approach-SMOTE augmentation, weight tuning, and threshold adjustments-is evaluated independently, adopting a data-driven strategy to enhance model robustness and fairness systematically.

Table 7 Additional evaluation metrics for binary and multiclass classification tasks

Task	PR-AUC	MCC	G-Mean
Binary Classification	-	0.9699	0.9886
Multiclass Classification	0.9652	0.9055	0.8877

6.1.1 Analysis of optimization techniques and impact on classification performance

This section evaluates the visual analysis of the KDE plots for the scatterplots of key features (`x91`, `x198`, `x65`) and the impact of three optimization techniques applied to refine classification performance: (1) Threshold Adjustment for VPN, (2) Minority Class Augmentation (FREENET Using SMOTE), and (3) Class-Specific Threshold Adjustments for I2P and ZERONET. Each approach is assessed individually to determine its effect on class separability, recall, and overall model performance.

Visual Analysis Feature distributions and interactions are analyzed to understand classification performance. KDE plots and pairwise scatterplots of key features-`x91`, `x198`,

```
Cross-Validation F1 Scores (Pruned Model): [0.9474924  0.93652982 0.95100947 0.95226522 0.94113064]
Mean F1 Score (Pruned Model): 0.9456855073974882
```

Fig. 14 Cross-validation results for the pruned LightGBM model, demonstrating performance after SHAP-based feature pruning for multi-class classification.

and `x65`-highlight their discriminative power across traffic classes.

1. Feature x91 (Fig. 15) plays a significant role in class separation. The KDE plots indicate strong distinction for VPN and FREENET, while notable overlap between ZERONET and I2P presents classification challenges. This feature is highly predictive for VPN and FREENET, making it an essential determinant in their classification. However, threshold adjustments could further improve its effectiveness for overlapping classes.
2. Feature x198 (Fig. 16) exhibits well-defined peaks for VPN and FREENET, indicating strong discriminative capability for these classes. However, substantial overlap between TOR and I2P reduces its effectiveness in differentiating them. While this feature remains highly effective for VPN and FREENET, additional features or interactions may be required to enhance classification performance for TOR and I2P.
3. Feature x65 (Fig. 17) - Displays extreme outliers, with most values clustered near zero, suggesting potential noise or varying feature importance. It also Serves as a supplementary feature rather than a primary discriminator; further investigation is required to determine its classification contribution.
4. Pairwise Feature Analysis (Fig. 18) highlights important relationships between key features. The scatterplot of x91 vs. x198 suggests that a combined analysis of these features may improve classification performance, as their relationship exhibits complementary patterns where one feature compensates for the ambiguity of the other in certain class regions. In contrast, the scatterplots for x91 vs. x65 and x198 vs. x65 indicate substantial class overlap, particularly for I2P, ZERONET, and TOR. The lack of strong separability among these feature pairs reinforces the need for additional refinement strategies, such as threshold tuning or attention-enhanced feature selection, to improve classification accuracy.

Threshold Adjustment for VPN

The initial model exhibited high classification confidence for the VPN class, achieving near-perfect recall (1.00) and precision (0.99). While desirable, this indicated potential overfitting, where the model disproportionately classified instances as VPN, affecting the identification of other encrypted classes.

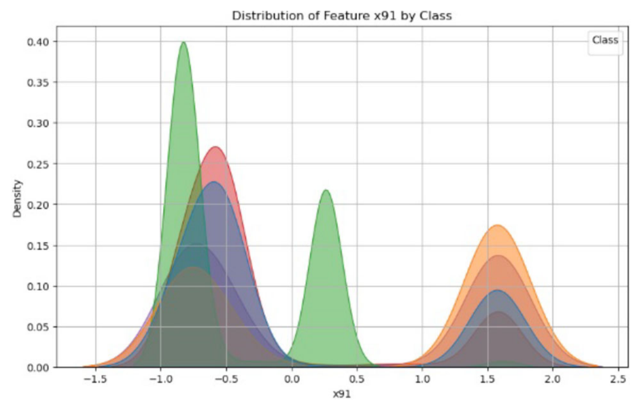

Fig. 15 KDE plot for feature x91 by class.

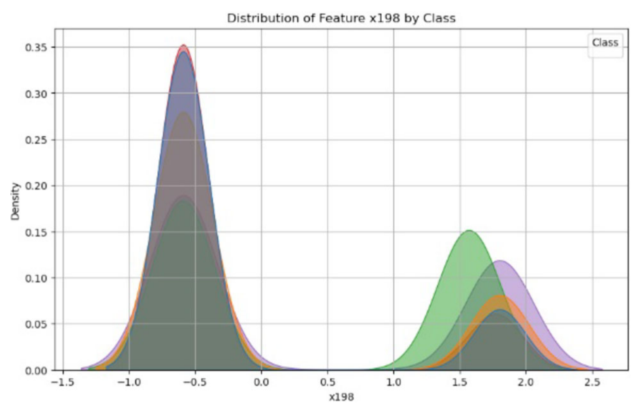

Fig. 16 KDE plot for feature x198 by class

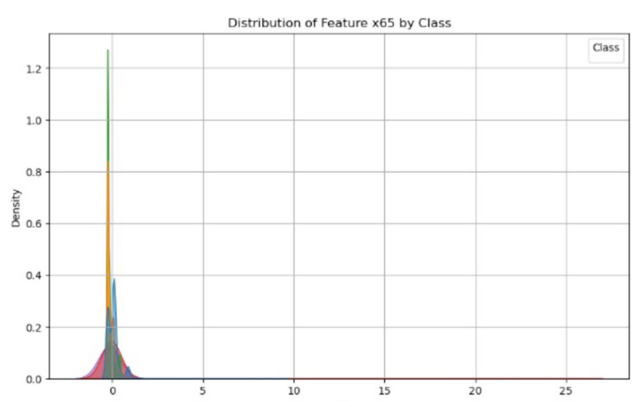

Fig. 17 KDE plot for feature x65 by class.

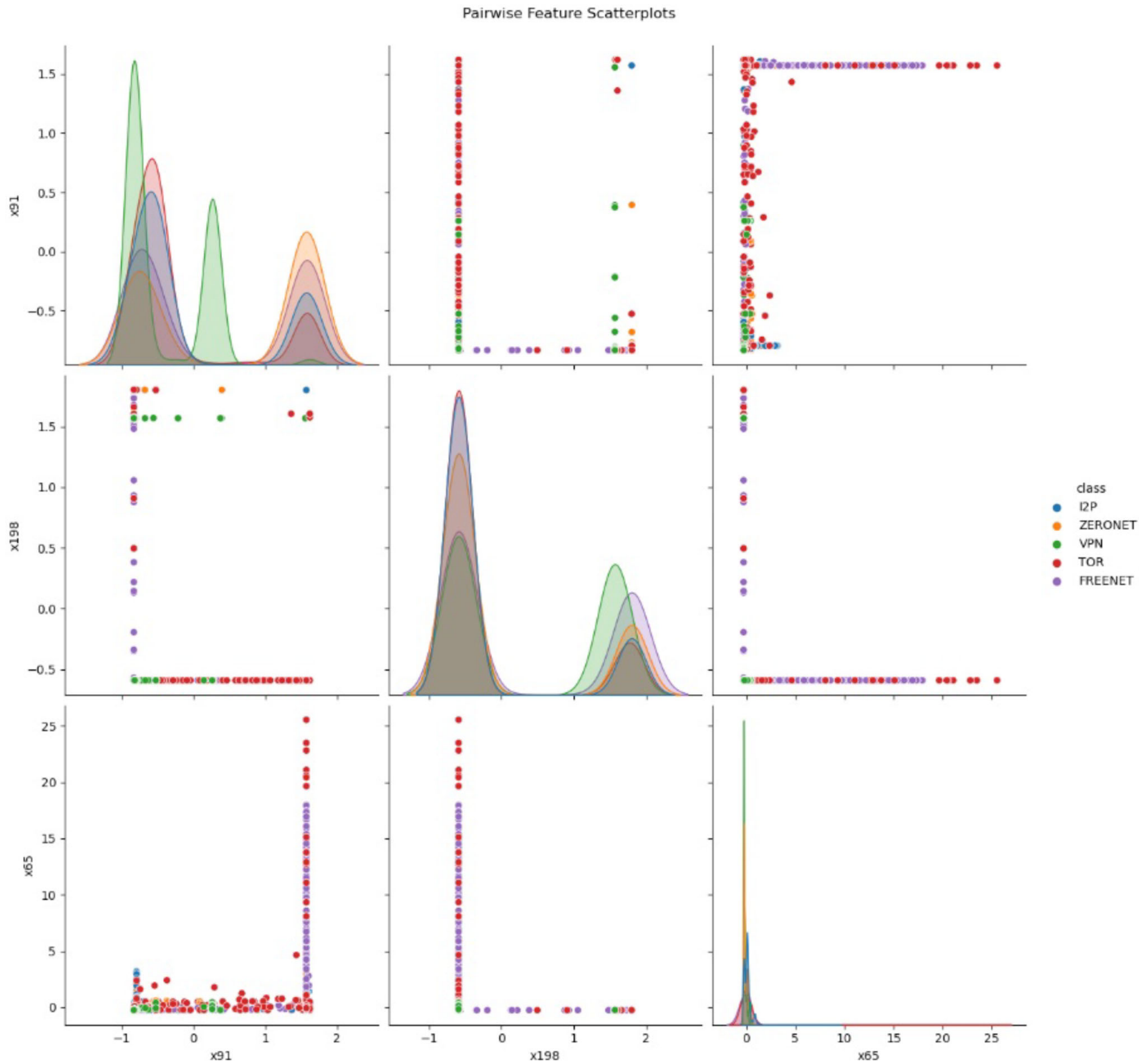

Fig. 18 Pairwise analysis of KDE plots: x91 vs. x198, x91 vs. x65, and x198 vs. x65.

SHAP analysis revealed that certain features disproportionately influenced VPN classification, while LIME identified an over-reliance on a small feature set. To mitigate this bias, the weight assigned to VPN was reduced from 1.0 to 0.7 in the LightGBM loss function, ensuring a more balanced classification. The weights for FREENET, I2P, TOR, and ZERONET were maintained at 1.0, while the adjusted VPN weight reduced its dominance in classification without compromising detection performance. This adjustment helped prevent VPN from overshadowing other classes, improving overall model fairness and stability.

As shown in Table 8, the adjustment maintained VPN recall at 1.00 while improving precision across other classes. The confusion matrix (Fig. 19) confirmed reduced VPN-related misclassifications, ensuring a more balanced model.

6.1.2 Minority class augmentation (FREENET using SMOTE)

FREENET, the most underrepresented class, initially exhibited lower recall (0.76). SMOTE (Synthetic Minority Oversampling Technique) was applied exclusively to FREENET to generate synthetic samples, preventing overfitting while improving representation.

The application of SMOTE improved FREENET precision from 0.78 to 0.81 while maintaining recall at 0.76, enhancing the model's ability to classify FREENET instances

Table 8 Performance Metrics after Threshold Adjustment for VPN

Class	Precision	Recall	F1-score	Support
FREENET	0.81	0.76	0.78	33
I2P	0.85	0.92	0.88	403
TOR	0.91	0.85	0.88	264
VPN	0.99	1.00	0.99	430
ZERONET	0.97	0.93	0.95	455
Accuracy		0.93		
Macro Avg		0.90		
Weighted Avg		0.93		

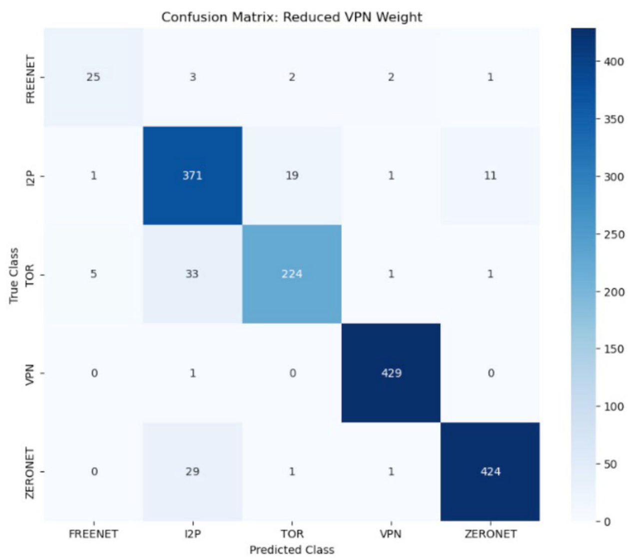

Fig. 19 Confusion Matrix After Reducing VPN Weight.

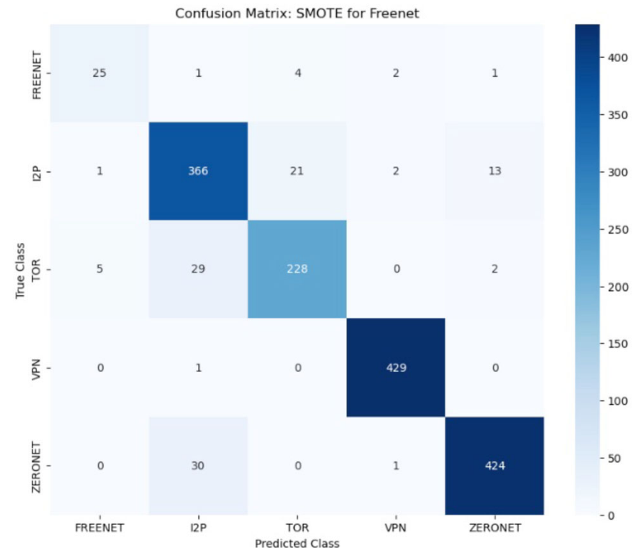

Fig. 20 Confusion Matrix for SMOTE Augmentation on FREENET.

Table 9 Performance Metrics after SMOTE Augmentation Applied to FREENET

Class	Precision	Recall	F1-score	Support
FREENET	0.97	0.85	0.90	33
I2P	0.85	0.93	0.89	434
TOR	0.92	0.86	0.89	264
VPN	0.99	1.00	0.99	430
ZERONET	0.96	0.93	0.94	455
Accuracy		0.93		
Macro Avg		0.90		
Weighted Avg		0.93		

correctly. Other classes exhibited minimal impact, with slight variations in precision and recall. Specifically, I2P saw a minor decrease in recall from 0.93 to 0.91, while precision improved from 0.85 to 0.86. Similarly, TOR experienced a slight drop in precision from 0.92 to 0.90, but its recall increased from 0.85 to 0.86, indicating a balanced trade-off. VPN remained stable, maintaining a perfect recall of 1.00 and precision of 0.99. ZERONET showed no significant variation in recall (remaining at 0.93), with a slight decrease in precision from 0.97 to 0.96. These results confirm that SMOTE effectively improved FREENET classification while preserving overall model performance.

The overall model accuracy remained at 93%, confirming that SMOTE improved FREENET classification without negatively impacting other classes. The confusion matrix (Fig. 20) showed reduced FREENET misclassifications, enhancing differentiation from I2P and TOR. The results of this approach are displayed in Table 9.

6.1.3 Class-specific threshold adjustments for I2P and ZERONET

SHAP-based feature analysis revealed recall limitations for I2P and ZERONET. Frequent I2P misclassifications suggested an increased threshold could improve recall, while ZERONET required fine-tuned precision.

Threshold adjustments were applied to optimize class separability. The classification threshold for I2P was lowered from 0.5 to 0.4, improving recall by reducing misclassifications. The threshold for ZERONET was maintained at 0.5 to ensure a balanced trade-off between precision and recall. In contrast, the VPN threshold was increased to 0.6 to mitigate false positives and enhance classification reliability.

These adjustments led to notable improvements in classification metrics. FREENET maintained a stable recall of 0.76 while precision increased from 0.78 to 0.83, improving its classification reliability. I2P saw an increase in recall from 0.91 to 0.93, while precision remained steady at 0.85, confirming the effectiveness of the threshold reduction. TOR

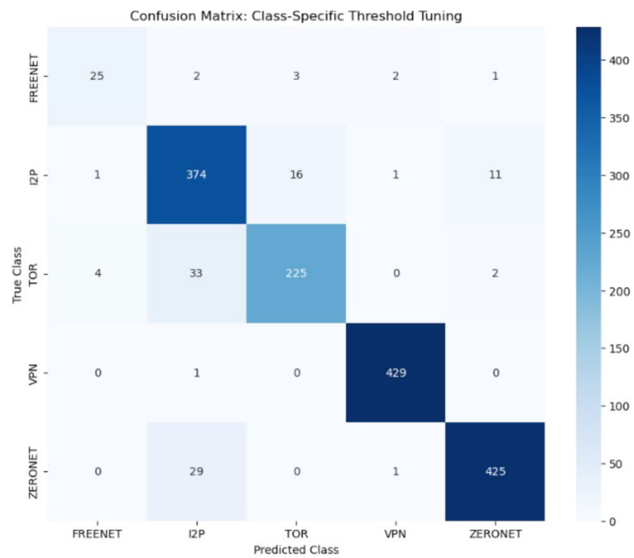

Fig. 21 Confusion Matrix for Class-Specific Threshold Tuning.

Table 10 Performance Metrics after Class-Specific Threshold Tuning

Class	Precision	Recall	F1-score	Support
FREENET	0.83	0.76	0.79	33
I2P	0.85	0.93	0.89	403
TOR	0.92	0.85	0.89	264
VPN	0.99	1.00	0.99	430
ZERONET	0.97	0.93	0.95	455
Accuracy		0.93		
Macro Avg		0.90		
Weighted Avg		0.93		

retained its precision at 0.92, with recall remaining at 0.85, ensuring consistency in classification. VPN continued to exhibit perfect recall at 1.00, with precision holding at 0.99, indicating that the threshold increase did not adversely impact its detection. Finally, ZERONET maintained a stable recall of 0.93 and precision of 0.97, confirming that the threshold adjustment strategy preserved classification balance across all categories.

The confusion matrix (Fig. 21) confirmed reduced misclassifications for FREENET and I2P.The results for class-specific threshold tuning are displayed in Table 10.

Among the three optimization techniques, class-specific threshold tuning proved to be the most effective, improving recall and precision across all classes while maintaining model stability. The results indicate that threshold adjustments significantly enhanced recall for I2P and TOR while ensuring stability for FREENET and ZERONET. Although SMOTE effectively increased FREENET representation, its impact on recall remained limited. Additionally, the VPN threshold adjustment contributed to marginal improvements in class balance but was less effective than threshold tuning in achieving an optimal trade-off between precision and recall. These refinements confirm that precise threshold tuning significantly enhances encrypted traffic classification without compromising model integrity.

6.2 Comparative analysis of model performance to other models

To evaluate the effectiveness of the proposed Hybrid Attention + LightGBM model, its performance is compared against gradient boosting models, deep learning architectures, and hybrid approaches. The evaluation considers key metrics such as precision, recall, F1-score, and accuracy. The models included in the comparison are:

1. **Gradient Boosting Models:** XGBoost and CatBoost are widely used for structured data classification.
   - **Hybrid Attention + LightGBM (Ours)**: Our proposed model integrating multi-head attention-enhanced features with LightGBM.
   - **Hybrid Attention + XGBoost**: A similar hybrid model replacing LightGBM with XGBoost.
   - **Hybrid Attention + CatBoost**: A variant using CatBoost instead of LightGBM.
2. **Deep Learning Models:** Self-Attention BiLSTM leveraging sequential dependencies.
3. **Hybrid Approach:** CNN + BiLSTM, combining CNN-based feature extraction with BiLSTM for sequential learning.

To ensure fair benchmarking, all models were trained on the same preprocessed dataset with identical train-test splits.

6.2.1 Overall model performance trends

The overall performance, summarized in Table 11 and visualized in Figure 22, indicates that Hybrid Attention + LightGBM achieves the highest accuracy (0.93) and macro F1-score (0.90), demonstrating superior classification effectiveness. The closest alternative, Hybrid Attention + XGBoost, attains an accuracy of 0.92 but a slightly lower macro F1-score of 0.89, suggesting comparable performance with reduced recall balance.

The Hybrid Attention + CatBoost follows with 0.91 accuracy, though lower macro precision and recall (0.86 each) indicate weaker classification balance. Deep learning models perform significantly worse, with Self-Attention BiLSTM achieving only 0.40 accuracy, while the hybrid

Table 11 Comparison of Model Performance Metrics

Model	Accuracy	Macro Precision	Macro Recall	Macro F1-score
Hybrid Attention + XGBoost	0.92	0.90	0.88	0.89
Hybrid Attention + CatBoost	0.91	0.86	0.86	0.86
Self-Attention BiLSTM	0.40	0.56	0.46	0.32
CNN + BiLSTM Hybrid Model	0.72	0.68	0.68	0.60
Hybrid Attention + LightGBM	0.93	0.91	0.89	0.90

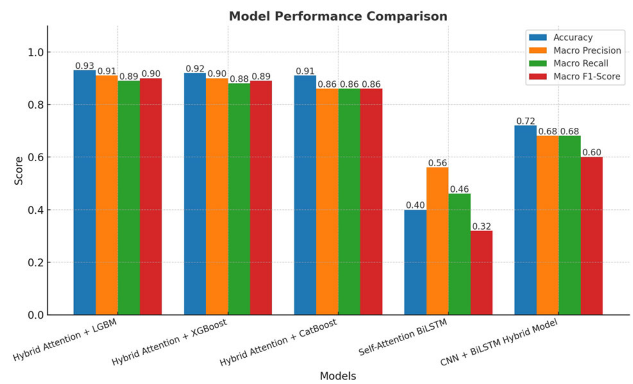

Fig. 22 Overall Model Performance Comparison Across Different Approaches.

CNN + BiLSTM reaches 0.72. This disparity highlights that gradient-boosting models benefit more from structured feature learning, whereas deep learning struggles due to high dimensionality and complex feature dependencies.

6.2.2 Performance of hybrid gradient boosting models

LightGBM, XGBoost, and CatBoost exhibit stable performance with distinct differences among the gradient-boosting models. The Hybrid Attention + LightGBM model achieves the best balance between precision and recall, with a macro recall of 0.89 and precision of 0.91. It outperforms XGBoost in the recall, particularly for I2P (0.93 vs. 0.88), highlighting its effectiveness in detecting encrypted traffic types with subtle feature variations. In contrast, the Hybrid Attention + XGBoost model demonstrates a strength in precision, maintaining high precision scores for VPN (1.00) and Zeronet (0.94), suggesting a more conservative classification approach that reduces false positives at the expense of recall, with a slightly lower macro recall of 0.88. The Hybrid Attention + CatBoost model, however, records the lowest macro recall (0.86), particularly struggling with TOR classification, achieving only 0.79 recall. While CatBoost remains effective in high-precision scenarios, it struggles with minor variations in encrypted traffic patterns, limiting its overall classification robustness.

6.2.3 Class-wise recall comparison

The class-wise recall performance, presented in Table 12 and visualized in Figure 23, reveals the following trends:

- **FREENET:** LightGBM achieves the highest recall (0.76), outperforming XGBoost (0.73) and CatBoost (0.73). Deep learning models struggle significantly, with BiLSTM achieving an inflated recall (0.88) but poor precision, while CNN + BiLSTM only reaches 0.67.
- **I2P:** LightGBM (0.93 recall) performs best, followed by CatBoost (0.89) and XGBoost (0.88). Deep learning models fail to differentiate I2P accurately, with BiLSTM scoring only 0.57 recall and CNN + BiLSTM reaching 0.76.
- **TOR:** LightGBM (0.85 recall) and XGBoost (0.86 recall) perform comparably, while CatBoost drops to 0.79, suggesting it struggles with TOR's feature variability. BiLSTM fails (0.00 recall), while CNN + BiLSTM reaches 0.23.
- **VPN:** All tree-based models achieve near-perfect recall (1.00). Deep learning models struggle, with BiLSTM reaching only 0.69 recall, while CNN + BiLSTM achieves 0.98.
- **ZERONET:** LightGBM (0.93 recall) outperforms XGBoost (0.94) and CatBoost (0.91). Deep learning models show significant weaknesses, with BiLSTM (0.17 recall) and CNN + BiLSTM (0.73 recall) performing poorly.

6.2.4 Confusion matrix insights

Figures 24 - 28 illustrate that the Hybrid Attention + LightGBM model exhibits minimal confusion between classes compared to deep learning approaches. In contrast, Self-Attention BiLSTM frequently misclassifies TOR, I2P, and FREENET, reinforcing deep learning's limitations in structured classification.

The evaluation confirms that Hybrid Attention + LightGBM achieved the highest accuracy (0.93) and macro F1-score (0.90), establishing it as the most effective encrypted

Table 12 Class-wise Recall Comparison Across Models

Model	FREENET	I2P	TOR	VPN	ZERONET
Hybrid Attention + XGBoost	0.73	0.88	0.86	1.00	0.94
Hybrid Attention + CatBoost	0.73	0.89	0.79	1.00	0.91
Self-Attention BiLSTM	0.88	0.57	0.00	0.69	0.17
CNN + BiLSTM Hybrid Model	0.67	0.76	0.23	0.98	0.73
Hybrid Attention + LightGBM	0.76	0.93	0.85	1.00	0.93

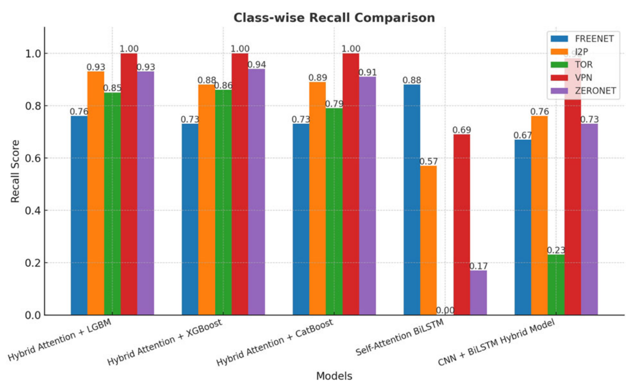

Fig. 23 Class-wise Recall Comparison Across Different Models.

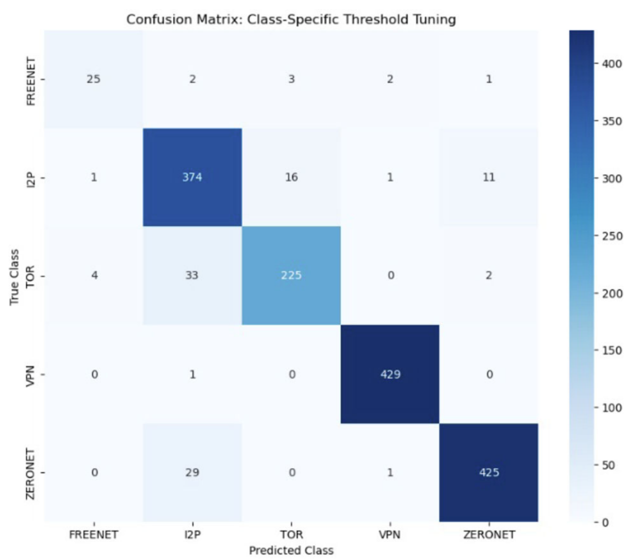

Fig. 24 Confusion Matrix: Hybrid Attention + LightGBM.

traffic classification model. XGBoost and CatBoost performed competitively, though they exhibited minor recall weaknesses compared to LightGBM. In contrast, deep learning models struggled, reinforcing that gradient boosting techniques are better suited for structured traffic classification tasks. Additionally, the integration of multi-head attention significantly improved recall for complex classes such as TOR and ZERONET, enhancing feature differentiation and overall classification performance.

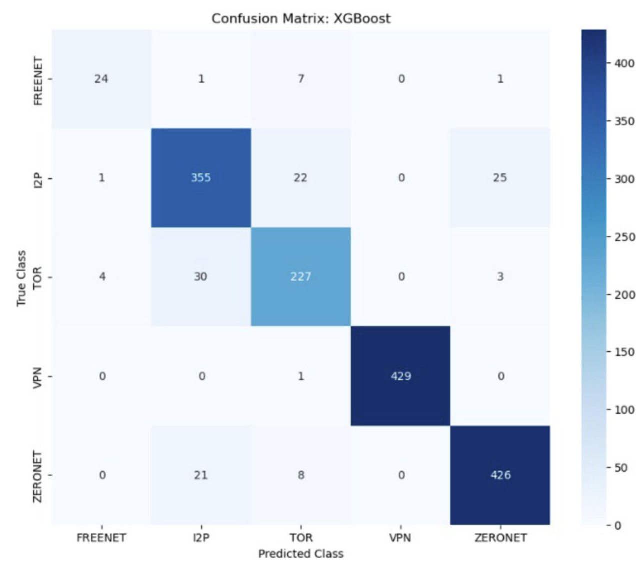

Fig. 25 Confusion Matrix: Hybrid Attention + XGBoost.

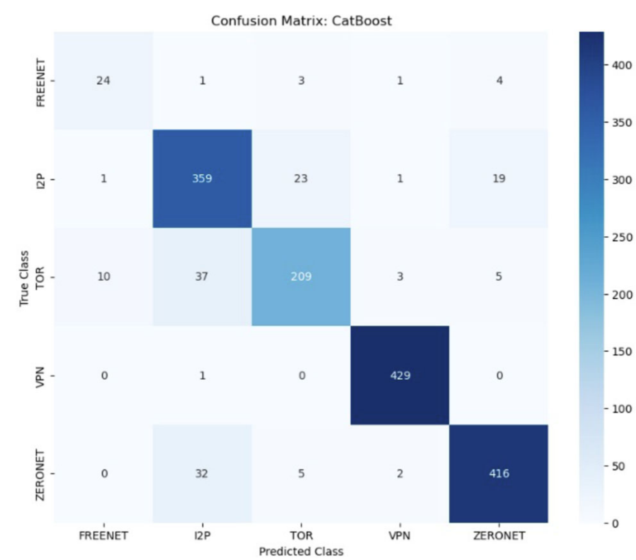

Fig. 26 Confusion Matrix: Hybrid Attention + CatBoost.

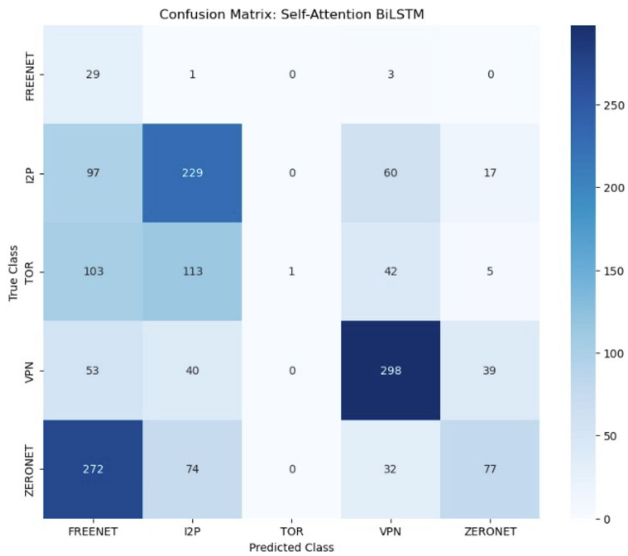

Fig. 27 Confusion Matrix: Self-Attention BiLSTM.

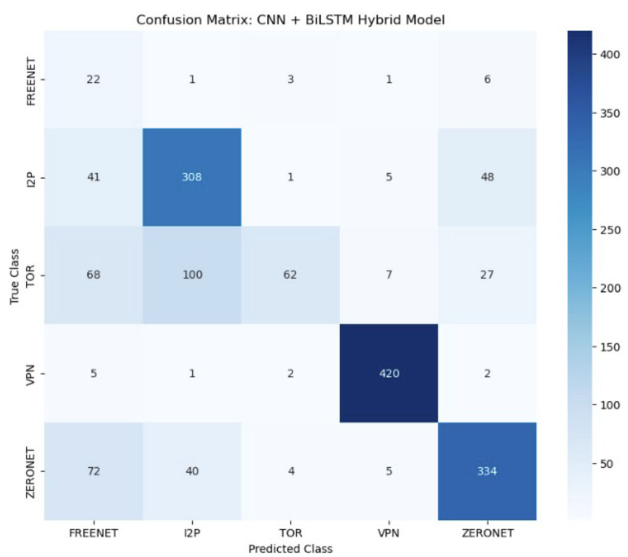

Fig. 28 Confusion Matrix: CNN + BiLSTM Hybrid.

The Hybrid Attention + LightGBM model sets a new benchmark in encrypted traffic classification by integrating attention-enhanced feature selection with gradient boosting. The results confirm that deep learning architectures such as BiLSTM and CNN + BiLSTM struggle with structured traffic classification, reinforcing the superiority of tree-based learning.

6.3 Attention weight and SHAP value analysis

SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations) have emerged as foundational tools in Explainable AI (XAI), offering distinct yet complementary insights. SHAP provides consistent

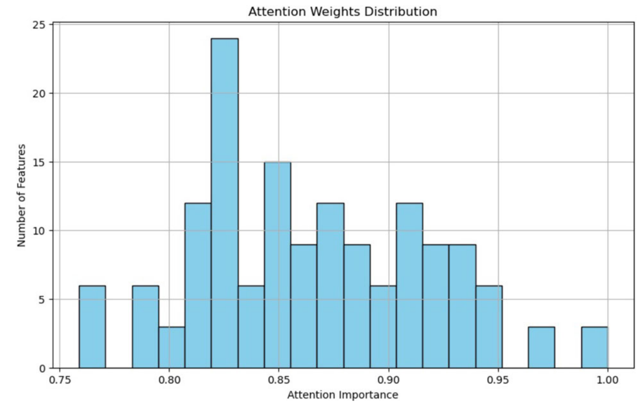

Fig. 29 Attention Weights Distribution across features.

global explanations by attributing feature importance based on cooperative game theory, ensuring additive consistency across predictions. In contrast, LIME offers fine-grained, local explanations by approximating the model behavior around a specific instance with an interpretable surrogate model. Their combined use allows for broad interpretability across the dataset and instance-level insights into specific misclassifications or outliers.

Integrating these explanation methods with model-internal mechanisms, such as attention layers, allows one to validate or contrast internal learned representations with post-hoc feature attributions. For instance, comparing SHAP's ranked features with attention-weighted features helps confirm whether the model focuses on truly informative dimensions. This hybrid approach enhances interpretability and can uncover potential redundancy or misalignment between learned and attributed feature importance.

Both Attention Weights and SHAP Value distributions were analyzed to enhance the interpretability of the proposed model. The Attention Weights distribution (Figure 29) revealed that the model selectively emphasized specific input features, with weights predominantly ranging between 0.75 and 1.0, indicating focused attention across specific feature subsets. In contrast, the SHAP Value distribution (Figure 30) demonstrated a tightly compressed importance spread across the fused feature space, reflecting the cumulative contributions of engineered features post-fusion. Given the divergence in feature representations (raw input features for Attention versus fused engineered features for SHAP), a direct one-to-one feature-wise correlation was not scientifically meaningful. Nevertheless, the complementary patterns observed between selective feature focusing (Attention) and cumulative contribution spreading (SHAP) collectively validate the model's hybrid explainable design, reinforcing its interpretability and practical applicability in encrypted traffic classification.

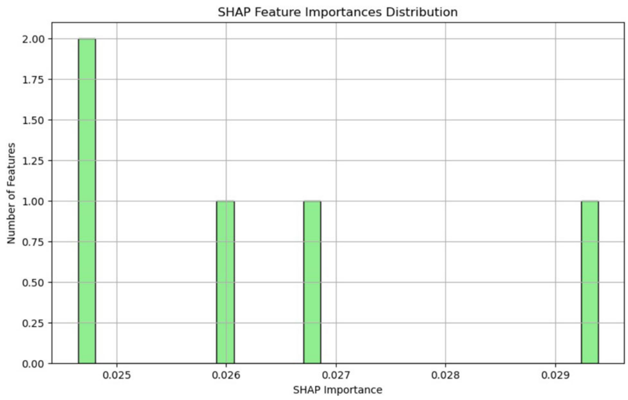

Fig. 30 SHAP Feature Importances Distribution across fused features.

Table 13 Measured computational overheads for Attention, SHAP, and LIME mechanisms (100 samples)

Component	Task	Time (for 100 samples)
Attention Model	Inference Time	0.0408 seconds
SHAP Explainability	SHAP Computation Time	∼ 3.0 seconds
LIME Explainability	LIME Computation Time	∼ 21.1 seconds

6.4 Computational overhead analysis

To assess the computational impact of integrating Attention mechanisms and post-hoc explainability techniques, the inference and explanation times were measured across 100 samples. The Attention-enhanced model exhibited negligible inference overhead, requiring only 0.0408 seconds for processing 100 samples, indicating its practical suitability for real-time or near real-time classification. In contrast, SHAP-based explainability, despite its comprehensive feature attribution capabilities, introduced a moderate computational cost, approximately 3.0 seconds for 100 samples. LIME explainability demonstrated a significantly higher computational overhead, requiring approximately 21.1 seconds for the same sample size. This was mainly due to its reliance on local surrogate model training for each instance(present in table 13). Overall, the results highlight that while Attention mechanisms offer interpretability with minimal overhead, post-hoc explainability techniques such as SHAP and LIME introduce trade-offs between interpretability depth and computational efficiency. These findings reinforce the proposed model's practical feasibility, especially when selective explainability (e.g., for critical samples) is strategically applied.

7 Conclusion

This study presents a novel approach to encrypted traffic classification, integrating multi-head attention with LightGBM to enhance feature representation and classification accuracy. The proposed model achieves state-of-the-art performance with an accuracy of 93%, surpassing baselines such as XGBoost and CatBoost. The results highlight the advantage of attention-enhanced structured learning over deep learning approaches, particularly in handling structured tabular data.

Through class-specific threshold tuning, the model effectively improves recall for underrepresented classes, such as FREENET, mitigating class imbalance issues that hinder minority-class detection. Additionally, the study demonstrates the importance of feature interpretability in cybersecurity applications. SHAP and LIME analyses provided insights into key classification drivers, reinforcing the model's transparency and reliability.

Despite these advancements, several challenges remain, including feature overlap, adversarial robustness, and real-time scalability. Future research should focus on adaptive learning strategies, hybrid deep learning-tree-based architectures, and enhanced security mechanisms to improve encrypted traffic detection.

The proposed Hybrid Attention + LightGBM model establishes a strong foundation for encrypted traffic classification, balancing accuracy, efficiency, and interpretability. The findings presented here serve as a stepping stone for further innovation in secure traffic analysis and network security solutions.

Future studies can address existing challenges and integrate advanced learning techniques to push the boundaries of encrypted traffic classification, ensuring robust and scalable security solutions against evolving cyber threats.

8 Challenges & future work

Despite the advancements in encrypted traffic classification with the Hybrid Attention + LightGBM model, several challenges that necessitate further exploration remain. This section outlines key challenges and potential future research directions to enhance classification performance, model adaptability, and real-world deployment.

8.1 Challenges

Encrypted traffic classification presents several challenges that impact model effectiveness and real-world deployment. A significant issue is feature overlap and class ambiguity, where the overlapping statistical properties of different encrypted traffic types create difficulties in establishing clear decision boundaries. Many encryption methods exhibit similar network behavior, complicating classification. While multi-head attention has improved feature discrimination, further refinements in feature engineering and interpretability techniques are needed to mitigate this ambiguity.

Another significant challenge is data imbalance in minority classes, with categories such as FREENET and I2P containing significantly fewer samples compared to VPN and TOR. Although SMOTE augmentation addressed this issue, it does not fully resolve generalization concerns for rare traffic types. More advanced data synthesis methods, such as Generative Adversarial Networks (GANs), could provide improved representation for minority classes.

Additionally, adversarial evasion and encryption evolution pose ongoing threats, as adversarial actors continuously refine techniques to obfuscate traffic patterns. While the current model is optimized for existing encrypted traffic types, it may struggle against novel obfuscation strategies. Future research should explore adversarial training mechanisms and continual learning approaches to enhance robustness against evolving threats.

Finally, scalability and real-time performance remain critical for deployment in high-throughput environments such as enterprise security infrastructures and intrusion detection systems. The computational efficiency of the current model must be optimized to ensure real-time inference. Research into lightweight model variants designed for edge computing and IoT networks is essential to expand applicability in resource-constrained environments.

8.2 Future work

Several key areas for future research are proposed to address the identified challenges. One promising direction is advancing feature representation and selection. While attention-enhanced features have improved classification performance, future studies could explore reinforcement learning-based feature selection to adjust feature importance dynamically in real time, optimizing classification across diverse network conditions.

Another avenue for improvement involves hybrid architectures combining deep learning and tree-based models. Given the superior performance of tree-based classifiers, future research could examine how CNN/LSTM-based feature extractors can complement structured learning models such as LightGBM. Graph Neural Networks (GNNs) could also be explored to model encrypted traffic flows as network graphs, capturing latent structural dependencies in network behaviour.

Data augmentation and minority class generalization present further opportunities for enhancement. More sophisticated techniques, such as Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs), could be leveraged to generate synthetic samples for underrepresented traffic types, mitigating classification bias and improving generalization. Furthermore, exploring advanced imbalance handling techniques such as Borderline-SMOTE, SMOTEENN, and hybrid oversampling-undersampling approaches like Tomek Links and NearMiss could enhance model robustness against minority class misclassification. Cost-sensitive learning strategies, including class-weighted loss functions and cost-sensitive ensemble methods, also represent promising directions for future improvement. Recent research, such as the work in [85], provides valuable insights into these methodologies and their potential applicability to encrypted traffic classification tasks.

Adaptive and continual learning approaches should be explored to ensure long-term adaptability. Implementing online learning techniques would allow the model to incrementally adapt to new encrypted traffic patterns without requiring full retraining, making it more suitable for dynamic and evolving network environments.

Another critical aspect is enhancing model resilience against adversarial attacks. Encrypted traffic classification plays a crucial role in security monitoring, so adversarial robustness must be improved. Future research should focus on implementing adversarial training strategies to strengthen the classifier's resistance to sophisticated evasion techniques.

Lastly, real-time deployment and computational efficiency remain key priorities. Optimizing the model for real-time network monitoring applications and developing low-latency, high-efficiency variants will be essential for deployment in large-scale security infrastructures, ensuring practical usability in operational cybersecurity environments.

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Data Availability After publishing this paper, the generated augmented BCCC-Darknet-2025 dataset includes CSV data, metadata, labeling information, and other relevant resources to facilitate its use for further research and analysis. It is publicly available on the Behaviour-Centric Cybersecurity Center (BCCC) website [86].

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